```
============================================================
  PARAMETER MANIFEST
============================================================

  Model           : Hubbard-Holstein (noise-model repository guide)
  Ansatz          : N/A (documentation artifact)
  Drive enabled   : False

  t (hopping)     : 0.0
  U (interaction) : 0.0
  dv (disorder)   : 0.0

  scope_root_modules: pipelines.exact_bench.hh_noise_hardware_validation,
    pipelines.exact_bench.hh_noise_robustness_seq_report, pipelines.exact_bench.hh_seq_transition_utils,
    pipelines.exact_bench.noise_oracle_runtime
  resolved_module_count: 34
  include_tests   : True
  include_docs    : True
  summary_json    : /Users/jakestrobel/Documents/Holstein_implementation/Holstein_test/docs/repo_guide_assets/
    repo_guide_summary.json

  Command:
    /opt/anaconda3/bin/python /Users/jakestrobel/Documents/Holstein_implementation/Holstein_test/pipelines/exa
    ct_bench/hh_noise_model_repo_guide.py --output-json /Users/jakestrobel/Documents/Holstein_implementation/H
    olstein_test/artifacts/json/hh_noise_model_repo_guide_index.json --output-pdf
    /Users/jakestrobel/Documents/Holstein_implementation/Holstein_test/docs/HH_noise_model_repo_guide.pdf

============================================================
```

**SECTION: HUMAN-READABLE FRONT MATTER**

# Human Guide Page 1/20: Purpose And Audience

Summary: This guide explains how noise behavior is implemented and validated in this repository.

- Primary audience: engineers extending/refactoring noise paths.
- Secondary audience: reviewers verifying contracts and regressions.
- Tertiary audience: automation/LLM tools consuming stable metadata.
- This front section is task-first; later sections remain exhaustive references.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 2)**

# Human Guide Page 2/20: One-Minute Noise Model Map

Summary: Four core modules define the noise stack end-to-end.

- noise_oracle_runtime: backend + estimator orchestration.
- hh_noise_hardware_validation: single-case noisy validation runner.
- hh_noise_robustness_seq_report: staged long-form robustness report.
- hh_seq_transition_utils: shared transition/pool/drive helper contracts.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 3)**

## Human Guide Page 3/20: How To Use This Document

Summary: Read the front pages first, then jump into sectioned references as needed.

- If you are changing behavior, read Refactor Playbook and tests first.
- If you are diagnosing a run, read Oracle + Validation contracts first.
- If you need complete symbol coverage, use appendix indices near the end.
- Section markers are stable for scripted PDF text-gate checks.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 4)**

# Human Guide Page 4/20: Mode Selection (Practical)

Summary: Pick noise mode by intent before tuning any optimization knobs.

- ideal: deterministic baseline and parity checks.
- shots: local stochastic estimator behavior without device model.
- aer_noise: backend-derived noise model emulation.
- runtime: real IBM Runtime backend execution path.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 5)**

# Human Guide Page 5/20: Inputs You Must Hold Constant

Summary: Noise-mode comparisons are only meaningful if physics settings are fixed.

- Keep L, t, U, dv, omega0, g_ep, n_ph_max unchanged across modes.
- Keep ordering, boundary, and ansatz consistent across comparisons.
- Keep trajectory grid knobs fixed (t_final, num_times, trotter_steps).
- Change one axis at a time to isolate estimator/backend effects.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 6)**

## Human Guide Page 6/20: Reproducibility Checklist

Summary: Seeds and repeats define stochastic reproducibility envelopes.

- Set seed/noise-seed and vqe/adapt seeds explicitly.
- Record shots, oracle_repeats, and oracle_aggregate every run.
- Capture exact command string and generated timestamp in artifacts.
- Persist JSON sidecar so comparisons are machine-auditable.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 7)**

# Human Guide Page 7/20: Backend Resolution Rules

Summary: Backend resolution behavior differs by mode and fake-backend controls.

- ideal mode uses statevector estimator path.
- shots/aer_noise prefer qiskit_aer estimator paths locally.
- runtime requires IBM credentials and backend-name availability.
- use-fake-backend drives deterministic fake-provider noise simulation.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 8)**

# Human Guide Page 8/20: OMP/SHM Failure Behavior

Summary: Local Aer can fail in restricted environments; fallback logic is explicit.

- OMP/SHM aborts are treated as environment-level failures, not physics bugs.
- allow-aer-fallback enables sampler_shots fallback for those failures.
- omp-shm-workaround applies environment controls for Aer startup resilience.
- Fallback details are emitted into backend/execution_fallback JSON fields.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 9)**

## Human Guide Page 9/20: Legacy Parity (Why It Matters)

Summary: Legacy parity ensures new noiseless-estimator paths preserve old physics outputs.

- Requires exact time-grid match, not just close trajectory lengths.
- Uses strict max_abs_delta gate against selected observables.
- Per-observable delta metrics are stored in legacy_parity payload blocks.
- Comparison plot output is available for quick visual forensic checks.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 10)**

# Human Guide Page 10/20: Validation Runner Outputs

Summary: Single-run validation emits a compact but comprehensive artifact set.

- JSON includes settings, noise_config, backend, vqe/adapt, trajectory.
- PDF is manifest-first with summary and trajectory/noisy-vs-ideal pages.
- Optional compare plot overlays new outputs with legacy references.
- Execution fallback metadata is explicit for postmortem review.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 11)**

# Human Guide Page 11/20: Reading Trajectory Delta Fields

Summary: Noise analysis hinges on measured-minus-ideal deltas per observable.

- energy_static_* and energy_total_* deltas track estimator/noise bias.
- site-0 occupation deltas expose spin-channel sensitivity.
- doublon/staggered deltas often surface nontrivial response drift.
- Always interpret delta panels with mode, shots, and repeats context.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 12)**

# Human Guide Page 12/20: Staged Robustness Flow

Summary: Long-form robustness report chains warm-start, ADAPT, and final VQE.

- Transition logic tracks |DeltaE| slope windows for stage handoff.
- Pool B is strict-union constrained and audited for provenance.
- Noiseless matrix benchmarks integrators before noisy overlays.
- Noisy matrix then adds per-mode estimator/backend effects.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 13)**

# Human Guide Page 13/20: Noisy Method Matrix

Summary: Noisy dynamics can be benchmarked per method and per mode.

- Default methods: suzuki2 and cfqm4; cfqm6 optional.
- Profiles include static and drive where configured.
- Success/failure is tracked per profile/method/mode branch.
- Mode availability failures are reported as structured diagnostics.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 14)**

# Human Guide Page 14/20: Benchmark Cost Metrics

Summary: Hardware-oriented proxy costs are tracked beside runtime totals.

- term_exp_count_total, pauli_rot_count_total, cx_proxy_total.
- sq_proxy_total, depth_proxy_total for additional complexity signals.
- wall_total_s and oracle_eval_s_total track runtime decomposition.
- Use proxy budgets as primary comparison axis; wall time is secondary.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 15)**

# Human Guide Page 15/20: Test Coverage Strategy

Summary: Tests enforce parser, fallback, parity, and benchmark-contract invariants.

- CLI tests verify defaulting and strict-minimum gate behavior.
- Oracle tests verify deterministic/fallback semantics and aggregations.
- Parity tests verify strict tolerance and time-grid mismatch handling.
- Benchmark tests verify method parsing, schema, and proxy-cost determinism.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 16)**

# Human Guide Page 16/20: Safe Refactor Boundaries

Summary: Certain contracts are non-negotiable when modifying noise code.

- Do not alter operator-core algebra modules for this workflow.
- Preserve exyz internal conventions and boundary conversions.
- Preserve manifest-first PDF contract for generated reports.
- Preserve strict legacy parity semantics and observable key meaning.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 17)**

# Human Guide Page 17/20: High-Risk Changes To Treat Carefully

Summary: A few change categories frequently introduce subtle regressions.

- Changing default mode flags without updating run-guide contracts.
- Mutating JSON key meaning while keeping key names unchanged.
- Breaking fallback diagnostics or suppressing failure reason fields.
- Changing trajectory-grid assumptions without parity/test updates.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 18)**

# Human Guide Page 18/20: Practical Extension Workflow

Summary: Use a repeatable workflow whenever adding a new noise feature.

- Add/adjust parser flags and defaults first.
- Thread feature through runner/oracle/report paths with clear payload keys.
- Add focused tests for parser + behavior + schema deltas.
- Update run-guide and regenerate this guide artifact.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 19)**

# Human Guide Page 19/20: Debugging Quickstart

Summary: Triage sequence for confusing noise behavior or mode failures.

- First inspect backend info + execution_fallback fields in JSON.
- Then inspect mode-specific unavailable reasons in robustness report.
- Then compare noisy-ideal deltas against repeats/shots settings.
- Finally run legacy parity anchor for noiseless-path confidence.

**SECTION: HUMAN-READABLE FRONT MATTER (cont. 20)**

# Human Guide Page 20/20: Quick Commands

Summary: Common command anchors for daily usage.

- Targeted tests: pytest -q test/test_hh_noise_*.py
- Validation runner: pipelines/exact_bench/hh_noise_hardware_validation.py
- Robustness runner: pipelines/exact_bench/hh_noise_robustness_seq_report.py
- This guide build: pipelines/shell/build_hh_noise_model_repo_guide.sh

```
SECTION: SCOPE + SOURCES

Noise-model full-repo guide scope is rooted at 4 fixed modules.
Scope expansion policy: roots + direct internal deps + inbound tests + doc references.

summary_json:
    /Users/jakestrobel/Documents/Holstein_implementation/Holstein_test/docs/repo_guide_assets/repo_guide_summary.js
    on
summary_found: True
fallback_ast_used: True
include_tests: True
include_docs: True

root_modules (4):
  - pipelines.exact_bench.hh_noise_hardware_validation
  - pipelines.exact_bench.hh_noise_robustness_seq_report
  - pipelines.exact_bench.hh_seq_transition_utils
  - pipelines.exact_bench.noise_oracle_runtime

resolved_modules (34):
  - pipelines.exact_bench.hh_noise_hardware_validation
  - pipelines.exact_bench.hh_noise_robustness_seq_report
  - pipelines.exact_bench.hh_seq_transition_utils
  - pipelines.exact_bench.noise_oracle_runtime
  - pipelines.hardcoded
  - src.quantum.drives_time_potential
  - src.quantum.hartree_fock_reference_state
  - src.quantum.hubbard_latex_python_pairs
  - src.quantum.operator_pools
  - src.quantum.pauli_polynomial_class
  - src.quantum.qubitization_module
  - src.quantum.time_propagation.cfqm_schemes
  - src.quantum.vqe_latex_python_pairs
  - test.test_adapt_vqe_integration
```

```
  - test.test_boson_unary
  - test.test_cfqm_acceptance
  - test.test_cfqm_propagator
  - test.test_cfqm_schemes
  - test.test_ed_crosscheck
  - test.test_energy_total_observables
  - test.test_exact_steps_multiplier
  - test.test_hh_drive_qop_builder
  - test.test_hh_noise_oracle_runtime
  - test.test_hh_noise_phase2_adapt
  - test.test_hh_noise_validation_cli
  - test.test_hh_pool_b_union
  - test.test_hh_seq_transition_logic
  - test.test_hubbard_adapt_ref_source
  - test.test_hubbard_holstein_ansatz
  - test.test_integration
  - test.test_pauli_polynomial_ops
  - test.test_time_potential_drive
  - test.test_trotter_hh_integration
  - test.test_vqe_hh_integration

inbound_test_modules (21):
  - test.test_adapt_vqe_integration
  - test.test_boson_unary
  - test.test_cfqm_acceptance
  - test.test_cfqm_propagator
  - test.test_cfqm_schemes
  - test.test_ed_crosscheck
  - test.test_energy_total_observables
  - test.test_exact_steps_multiplier
  - test.test_hh_drive_qop_builder
  - test.test_hh_noise_oracle_runtime
  - test.test_hh_noise_phase2_adapt
  - test.test_hh_noise_validation_cli
```

```
  - test.test_hh_pool_b_union
  - test.test_hh_seq_transition_logic
  - test.test_hubbard_adapt_ref_source
  - test.test_hubbard_holstein_ansatz
  - test.test_integration
  - test.test_pauli_polynomial_ops
  - test.test_time_potential_drive
  - test.test_trotter_hh_integration
  - test.test_vqe_hh_integration

doc_reference_files (5):
  - README.md
  - docs/LLM_RESEARCH_CONTEXT.md
  - docs/repo_implementation_guide.md
  - pipelines/exact_bench/README.md
  - pipelines/run_guide.md
```

SECTION: ARCHITECTURE MAP | Architecture modules in resolved noise scope

| path | module | role | symbols | cli_flags |
|---|---|---|---|---|
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | single-run HH noisy validation | 28 | 58 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | staged HH robustness + long report | 65 | 73 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | Pool-B/transition helper kernel | 10 | 0 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | shared expectation oracle | 29 | 0 |
| pipelines/hardcoded.py | pipelines.hardcoded | direct dependency | 0 | 0 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | direct dependency | 10 | 0 |
| src/quantum/hartree_fock_reference_state.p | src.quantum.hartree_fock_reference_state | direct dependency | 9 | 0 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | direct dependency | 41 | 0 |
| src/quantum/operator_pools.py | src.quantum.operator_pools | direct dependency | 0 | 0 |
| src/quantum/pauli_polynomial_class.py | src.quantum.pauli_polynomial_class | direct dependency | 3 | 0 |
| src/quantum/qubitization_module.py | src.quantum.qubitization_module | direct dependency | 1 | 0 |
| src/quantum/time_propagation/cfqm_schemes | src.quantum.time_propagation.cfqm_scheme | direct dependency | 9 | 0 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | direct dependency | 38 | 0 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | test coverage | 12 | 0 |
| test/test_boson_unary.py | test.test_boson_unary | test coverage | 15 | 0 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | test coverage | 7 | 0 |
| test/test_cfqm_propagator.py | test.test_cfqm_propagator | test coverage | 2 | 0 |
| test/test_cfqm_schemes.py | test.test_cfqm_schemes | test coverage | 2 | 0 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | test coverage | 11 | 0 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | test coverage | 10 | 0 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | test coverage | 9 | 0 |
| test/test_hh_drive_qop_builder.py | test.test_hh_drive_qop_builder | test coverage | 3 | 0 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | test coverage | 7 | 0 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | test coverage | 9 | 0 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | test coverage | 6 | 0 |
| test/test_hh_pool_b_union.py | test.test_hh_pool_b_union | test coverage | 2 | 0 |

SECTION: ARCHITECTURE MAP | Architecture modules in resolved noise scope (cont. 2)

| path | module | role | symbols | cli_flags |
|---|---|---|---|---|
| test/test_hh_seq_transition_logic.py | test.test_hh_seq_transition_logic | test coverage | 2 | 0 |
| test/test_hubbard_adapt_ref_source.py | test.test_hubbard_adapt_ref_source | test coverage | 5 | 0 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | test coverage | 10 | 0 |
| test/test_integration.py | test.test_integration | test coverage | 8 | 0 |
| test/test_pauli_polynomial_ops.py | test.test_pauli_polynomial_ops | test coverage | 4 | 0 |
| test/test_time_potential_drive.py | test.test_time_potential_drive | test coverage | 6 | 0 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | test coverage | 21 | 0 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | test coverage | 24 | 0 |

SECTION: ARCHITECTURE MAP | Internal import edges (scope-focused)

| src | dst |
|---|---|
| pipelines.exact_bench.cfqm_vs_suzuki_qproc_proxy_benchmark | src.quantum.drives_time_potential |
| pipelines.exact_bench.cfqm_vs_suzuki_qproc_proxy_benchmark | src.quantum.time_propagation.cfqm_schemes |
| pipelines.exact_bench.cross_check_suite | src.quantum.hartree_fock_reference_state |
| pipelines.exact_bench.cross_check_suite | src.quantum.hubbard_latex_python_pairs |
| pipelines.exact_bench.cross_check_suite | src.quantum.pauli_polynomial_class |
| pipelines.exact_bench.cross_check_suite | src.quantum.vqe_latex_python_pairs |
| pipelines.exact_bench.hh_noise_hardware_validation | pipelines.exact_bench.noise_oracle_runtime |
| pipelines.exact_bench.hh_noise_hardware_validation | src.quantum.hartree_fock_reference_state |
| pipelines.exact_bench.hh_noise_hardware_validation | src.quantum.hubbard_latex_python_pairs |
| pipelines.exact_bench.hh_noise_hardware_validation | src.quantum.pauli_polynomial_class |
| pipelines.exact_bench.hh_noise_hardware_validation | src.quantum.qubitization_module |
| pipelines.exact_bench.hh_noise_hardware_validation | src.quantum.vqe_latex_python_pairs |
| pipelines.exact_bench.hh_noise_robustness_seq_report | pipelines.exact_bench.hh_seq_transition_utils |
| pipelines.exact_bench.hh_noise_robustness_seq_report | pipelines.exact_bench.noise_oracle_runtime |
| pipelines.exact_bench.hh_noise_robustness_seq_report | pipelines.hardcoded |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.drives_time_potential |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.hartree_fock_reference_state |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.hubbard_latex_python_pairs |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.operator_pools |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.pauli_polynomial_class |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.qubitization_module |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.time_propagation.cfqm_schemes |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.vqe_latex_python_pairs |
| pipelines.exact_bench.hh_seq_transition_utils | src.quantum.vqe_latex_python_pairs |
| pipelines.exact_bench.l3_hh_exp_fidelity_wrapper | src.quantum.hubbard_latex_python_pairs |
| pipelines.exact_bench.l3_hh_exp_fidelity_wrapper | src.quantum.operator_pools |

SECTION: ARCHITECTURE MAP | Internal import edges (scope-focused) (cont. 2)

| src | dst |
|---|---|
| pipelines.exact_bench.l3_hh_exp_fidelity_wrapper | src.quantum.vqe_latex_python_pairs |
| pipelines.exact_bench.overnight_l3_hh_four_method_benchmark | src.quantum.hubbard_latex_python_pairs |
| pipelines.exact_bench.overnight_l3_hh_four_method_benchmark | src.quantum.operator_pools |
| pipelines.exact_bench.overnight_l3_hh_four_method_benchmark | src.quantum.vqe_latex_python_pairs |
| pipelines.hardcoded.adapt_pipeline | src.quantum.hartree_fock_reference_state |
| pipelines.hardcoded.adapt_pipeline | src.quantum.hubbard_latex_python_pairs |
| pipelines.hardcoded.adapt_pipeline | src.quantum.operator_pools |
| pipelines.hardcoded.adapt_pipeline | src.quantum.pauli_polynomial_class |
| pipelines.hardcoded.adapt_pipeline | src.quantum.vqe_latex_python_pairs |
| pipelines.hardcoded.bench_expm | src.quantum.drives_time_potential |
| pipelines.hardcoded.bench_expm | src.quantum.hubbard_latex_python_pairs |
| pipelines.hardcoded.hubbard_pipeline | pipelines.hardcoded |
| pipelines.hardcoded.hubbard_pipeline | src.quantum.drives_time_potential |
| pipelines.hardcoded.hubbard_pipeline | src.quantum.hartree_fock_reference_state |
| pipelines.hardcoded.hubbard_pipeline | src.quantum.hubbard_latex_python_pairs |
| pipelines.hardcoded.hubbard_pipeline | src.quantum.time_propagation.cfqm_schemes |
| pipelines.qiskit_archive.compare_hc_vs_qk | src.quantum.drives_time_potential |
| pipelines.qiskit_archive.qiskit_baseline | src.quantum.drives_time_potential |
| pipelines.qiskit_archive.qiskit_baseline | src.quantum.hartree_fock_reference_state |
| src.quantum.drives_time_potential | src.quantum.hubbard_latex_python_pairs |
| src.quantum.ed_hubbard_holstein | src.quantum.hubbard_latex_python_pairs |
| src.quantum.hartree_fock_reference_state | src.quantum.hubbard_latex_python_pairs |
| src.quantum.hubbard_latex_python_pairs | src.quantum.pauli_polynomial_class |
| src.quantum.hubbard_latex_python_pairs | src.quantum.qubitization_module |
| src.quantum.operator_pools.polaron_paop | src.quantum.hubbard_latex_python_pairs |
| src.quantum.operator_pools.polaron_paop | src.quantum.pauli_polynomial_class |

SECTION: ARCHITECTURE MAP | Internal import edges (scope-focused) (cont. 3)

| src | dst |
|---|---|
| src.quantum.operator_pools.polaron_paop | src.quantum.qubitization_module |
| src.quantum.pauli_polynomial_class | src.quantum.qubitization_module |
| src.quantum.pauli_words | src.quantum.qubitization_module |
| src.quantum.qubitization_module | src.quantum.pauli_letters_module |
| src.quantum.vqe_latex_python_pairs | hartree_fock_reference_state |
| src.quantum.vqe_latex_python_pairs | src.quantum.compiled_polynomial |
| src.quantum.vqe_latex_python_pairs | src.quantum.hartree_fock_reference_state |
| src.quantum.vqe_latex_python_pairs | src.quantum.hubbard_latex_python_pairs |
| src.quantum.vqe_latex_python_pairs | src.quantum.pauli_polynomial_class |
| src.quantum.vqe_latex_python_pairs | src.quantum.pauli_words |
| test.test_adapt_vqe_integration | src.quantum.hubbard_latex_python_pairs |
| test.test_adapt_vqe_integration | src.quantum.operator_pools |
| test.test_adapt_vqe_integration | src.quantum.vqe_latex_python_pairs |
| test.test_boson_unary | src.quantum.hartree_fock_reference_state |
| test.test_boson_unary | src.quantum.hubbard_latex_python_pairs |
| test.test_boson_unary | src.quantum.vqe_latex_python_pairs |
| test.test_cfqm_acceptance | pipelines.hardcoded.hubbard_pipeline |
| test.test_cfqm_acceptance | src.quantum.drives_time_potential |
| test.test_cfqm_acceptance | src.quantum.hubbard_latex_python_pairs |
| test.test_cfqm_acceptance | src.quantum.time_propagation.cfqm_propagator |
| test.test_cfqm_acceptance | src.quantum.time_propagation.cfqm_schemes |
| test.test_cfqm_propagator | src.quantum.time_propagation.cfqm_propagator |
| test.test_cfqm_propagator | src.quantum.time_propagation.cfqm_schemes |
| test.test_cfqm_schemes | src.quantum.time_propagation.cfqm_schemes |
| test.test_ed_crosscheck | src.quantum.ed_hubbard_holstein |
| test.test_ed_crosscheck | src.quantum.hubbard_latex_python_pairs |

SECTION: ARCHITECTURE MAP | Internal import edges (scope-focused) (cont. 4)

| src | dst |
|---|---|
| test.test_ed_crosscheck | src.quantum.vqe_latex_python_pairs |
| test.test_energy_total_observables | pipelines.hardcoded.hubbard_pipeline |
| test.test_energy_total_observables | src.quantum.drives_time_potential |
| test.test_energy_total_observables | src.quantum.hartree_fock_reference_state |
| test.test_energy_total_observables | src.quantum.hubbard_latex_python_pairs |
| test.test_exact_steps_multiplier | pipelines.hardcoded.hubbard_pipeline |
| test.test_exact_steps_multiplier | pipelines.qiskit_archive.compare_hc_vs_qk |
| test.test_exact_steps_multiplier | src.quantum.drives_time_potential |
| test.test_hh_drive_qop_builder | pipelines.exact_bench.hh_seq_transition_utils |
| test.test_hh_noise_oracle_runtime | pipelines.exact_bench.noise_oracle_runtime |
| test.test_hh_noise_oracle_runtime | src.quantum.hartree_fock_reference_state |
| test.test_hh_noise_oracle_runtime | src.quantum.pauli_polynomial_class |
| test.test_hh_noise_oracle_runtime | src.quantum.qubitization_module |
| test.test_hh_noise_oracle_runtime | src.quantum.vqe_latex_python_pairs |
| test.test_hh_noise_phase2_adapt | pipelines.exact_bench.hh_noise_hardware_validation |
| test.test_hh_noise_validation_cli | pipelines.exact_bench.hh_noise_hardware_validation |
| test.test_hh_pool_b_union | pipelines.exact_bench.hh_seq_transition_utils |
| test.test_hh_pool_b_union | src.quantum.pauli_polynomial_class |
| test.test_hh_pool_b_union | src.quantum.qubitization_module |
| test.test_hh_pool_b_union | src.quantum.vqe_latex_python_pairs |
| test.test_hh_seq_transition_logic | pipelines.exact_bench.hh_seq_transition_utils |
| test.test_hubbard_adapt_ref_source | pipelines.hardcoded.hubbard_pipeline |
| test.test_hubbard_adapt_ref_source | src.quantum.hubbard_latex_python_pairs |
| test.test_hubbard_holstein_ansatz | src.quantum.hartree_fock_reference_state |
| test.test_hubbard_holstein_ansatz | src.quantum.hubbard_latex_python_pairs |
| test.test_hubbard_holstein_ansatz | src.quantum.vqe_latex_python_pairs |

SECTION: ARCHITECTURE MAP | Internal import edges (scope-focused) (cont. 5)

| src | dst |
|---|---|
| test.test_integration | src.quantum.hartree_fock_reference_state |
| test.test_integration | src.quantum.hubbard_latex_python_pairs |
| test.test_integration | src.quantum.pauli_letters_module |
| test.test_integration | src.quantum.pauli_polynomial_class |
| test.test_integration | src.quantum.pauli_words |
| test.test_pauli_polynomial_ops | src.quantum.pauli_polynomial_class |
| test.test_pauli_polynomial_ops | src.quantum.qubitization_module |
| test.test_time_potential_drive | pipelines.hardcoded.hubbard_pipeline |
| test.test_time_potential_drive | src.quantum.drives_time_potential |
| test.test_time_potential_drive | src.quantum.hubbard_latex_python_pairs |
| test.test_trotter_hh_integration | pipelines.hardcoded.hubbard_pipeline |
| test.test_trotter_hh_integration | pipelines.qiskit_archive.compare_hc_vs_qk |
| test.test_trotter_hh_integration | pipelines.qiskit_archive.qiskit_baseline |
| test.test_trotter_hh_integration | src.quantum.hartree_fock_reference_state |
| test.test_trotter_hh_integration | src.quantum.hubbard_latex_python_pairs |
| test.test_trotter_hh_integration | src.quantum.vqe_latex_python_pairs |
| test.test_vqe_hh_integration | src.quantum.hartree_fock_reference_state |
| test.test_vqe_hh_integration | src.quantum.hubbard_latex_python_pairs |
| test.test_vqe_hh_integration | src.quantum.vqe_latex_python_pairs |

```
SECTION: ORACLE CONTRACT

Module: pipelines.exact_bench.noise_oracle_runtime
Core execution modes: ideal, shots, aer_noise, runtime.
Fallback contract: OMP/SHM failures in Aer paths may route to sampler_shots fallback when enabled.
Runtime contract: runtime mode requires IBM Runtime backend resolution and credentials.

Key behavior surfaces:
  - OracleConfig / OracleEstimate / NoiseBackendInfo
  - ExpectationOracle.evaluate aggregate policy (mean|median over repeats)
  - _resolve_noise_backend, _build_estimator, _run_estimator_job
  - _looks_like_openmp_shm_abort and fallback activation

Relevant validation CLI flags (from hh_noise_hardware_validation):
  - --allow-aer-fallback
  - --backend-name
  - --compare-observables
  - --legacy-parity-tol
  - --legacy-reference-json
  - --no-allow-aer-fallback
  - --no-omp-shm-workaround
  - --noise-mode
  - --omp-shm-workaround
  - --oracle-aggregate
  - --oracle-repeats
  - --output-compare-plot
  - --shots
  - --use-fake-backend
  - --vqe-method
```

```
SECTION: VALIDATION PIPELINE CONTRACT

Module: pipelines.exact_bench.hh_noise_hardware_validation
Contract summary:
  1. Build HH/Hubbard Hamiltonian + ansatz/reference state.
  2. Instantiate shared noisy and ideal oracles.
  3. Optional ADAPT stage under noisy oracle.
  4. Optional VQE stage under noisy oracle.
  5. Optional Trotter trajectory with noisy-vs-ideal deltas.
  6. Optional legacy parity audit against locked reference JSON.
  7. Emit JSON + manifest-first PDF (+ optional compare plot).

Primary JSON areas:
  - settings, noise_config, backend, execution_fallback
  - vqe, adapt, trajectory, summary
  - legacy_parity (when enabled)

Legacy parity strictness:
  - exact time-grid match required
  - max_abs_delta <= legacy_parity_tol (default 1e-10)
```

```
SECTION: ROBUSTNESS REPORT CONTRACT

Module: pipelines.exact_bench.hh_noise_robustness_seq_report
Staged flow:
  1. warm-start HVA
  2. ADAPT Pool-B strict union (UCCSD_lifted + HVA + PAOP_full)
  3. final seeded conventional VQE
  4. noiseless matrix (suzuki2/magnus2/cfqm4/cfqm6)
  5. noisy method matrix per profile and mode

Noisy matrix defaults:
  - noise_modes: ideal,shots,aer_noise
  - noisy_methods: suzuki2,cfqm4

Benchmark schema (per successful method+mode):
  - benchmark_cost.{term_exp_count_total,pauli_rot_count_total,cx_proxy_total,sq_proxy_total,depth_proxy_total}
  - benchmark_runtime.{wall_total_s,circuit_build_s_total,oracle_eval_s_total,oracle_calls_total}

Output contract:
  - long manifest-first PDF with section markers and caption strips
  - JSON with equation_registry and plot_contracts for auditability
```

SECTION: TEST + DOC COVERAGE

Inbound tests document and enforce noise-stack contracts.
Doc references capture run-guide + implementation-guide contract anchors.

test_anchor_count: 67
doc_reference_count: 338

SECTION: TEST + DOC COVERAGE | Inbound test anchors (function-level)

| path | module | test_function | line |
|---|---|---|---|
| test/test_boson_unary.py | test.test_boson_unary | _term_strings | 52 |
| test/test_boson_unary.py | test.test_boson_unary | _pauli_letters_in_poly | 57 |
| test/test_boson_unary.py | test.test_boson_unary | _is_hermitian | 67 |
| test/test_boson_unary.py | test.test_boson_unary | _one_hot_projector | 71 |
| test/test_boson_unary.py | test.test_boson_unary | _binary_physical_projector | 107 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | _run_cfqm_total | 29 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | _estimate_state_order_from_infidelity | 63 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | test_cfqm4_static_regression_against_exact_expm | 80 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | test_cfqm4_a0_safe_test_invariance_via_real_drive_provid | 106 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | test_manufactured_one_qubit_order_confirmation | 166 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | test_small_hh_cfqm4_converges_toward_fine_piecewise_ex | 231 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | _sector_eigenvalues_pauli | 48 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | _sector_eigenvalues_ed | 89 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | _build_L2_blocked_hamiltonian | 43 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | _build_drive | 55 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | _run_short_trajectory | 71 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | _import_evolve_piecewise_exact | 63 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | _exact_psi_sin_z | 124 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | _run_piecewise_exact | 133 |
| test/test_hh_drive_qop_builder.py | test.test_hh_drive_qop_builder | _coeff_map | 8 |
| test/test_hh_drive_qop_builder.py | test.test_hh_drive_qop_builder | test_drive_qop_builder_combines_static_and_drive_coefficie | 15 |
| test/test_hh_drive_qop_builder.py | test.test_hh_drive_qop_builder | test_drive_qop_builder_identity_fallback_when_all_terms_ca | 28 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | test_pauli_poly_to_sparse_pauli_op_preserves_exyz_to_ixyz_m | 22 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | test_ansatz_to_circuit_matches_prepare_state_for_hh_termwise_s | 37 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | test_ideal_oracle_matches_statevector_expectation | 76 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | test_shots_oracle_standard_error_improves_with_more_rep | 91 |

SECTION: TEST + DOC COVERAGE | Inbound test anchors (function-level) (cont. 2)

| path | module | test_function | line |
|---|---|---|---|
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | test_aer_noise_mode_deterministic_with_fixed_seed_and_fake_ | 127 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | test_forced_aer_failure_triggers_sampler_fallback | 154 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | test_sampler_fallback_deterministic_with_fixed_seed | 180 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | test_parse_args_phase2_adapt_flags_present | 16 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | test_build_adapt_pool_hh_hva_nonempty | 43 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | test_build_adapt_pool_hubbard_uccsd_nonempty | 51 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | _legacy_ref_fixture | 70 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | _new_traj_fixture | 81 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | test_legacy_parity_exact_match_passes_strict_tol | 100 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | test_legacy_parity_tiny_mismatch_fails_strict_tol | 113 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | test_legacy_parity_time_grid_mismatch_fails_with_reaso | 126 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | test_legacy_comparison_plot_helper_smoke | 138 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | test_hh_defaults_applied_from_minimum_table_l2_nph1 | 11 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | test_hh_under_minimum_rejected_without_smoke_flag | 21 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | test_hh_under_minimum_allowed_with_smoke_flag | 40 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | test_hubbard_under_minimum_rejected_without_smoke_f | 64 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | test_cli_parses_fallback_flags | 87 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | test_cli_parses_legacy_parity_flags | 100 |
| test/test_hh_pool_b_union.py | test.test_hh_pool_b_union | _single_term_op | 9 |
| test/test_hh_pool_b_union.py | test.test_hh_pool_b_union | test_pool_b_strict_union_dedups_and_tracks_presence | 15 |
| test/test_hh_seq_transition_logic.py | test.test_hh_seq_transition_logic | test_windowed_slope_transition_triggers_after_patience | 11 |
| test/test_hh_seq_transition_logic.py | test.test_hh_seq_transition_logic | test_windowed_slope_transition_does_not_trigger_when_still_de | 32 |
| test/test_hubbard_adapt_ref_source.py | test.test_hubbard_adapt_ref_source | _hh_state_dim | 22 |
| test/test_hubbard_adapt_ref_source.py | test.test_hubbard_adapt_ref_source | _basis0 | 28 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | n_ph_max | 39 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | L | 44 |

SECTION: TEST + DOC COVERAGE | Inbound test anchors (function-level) (cont. 3)

| path | module | test_function | line |
|---|---|---|---|
| test/test_integration.py | test.test_integration | _add_repo_paths | 13 |
| test/test_integration.py | test.test_integration | _run_import_check | 22 |
| test/test_integration.py | test.test_integration | _import_pauli_letters | 35 |
| test/test_integration.py | test.test_integration | _import_pauli_term | 44 |
| test/test_integration.py | test.test_integration | _import_pauli_polynomial | 50 |
| test/test_integration.py | test.test_integration | _import_hubbard_helpers | 64 |
| test/test_integration.py | test.test_integration | _import_hf_helpers | 78 |
| test/test_integration.py | test.test_integration | main | 90 |
| test/test_pauli_polynomial_ops.py | test.test_pauli_polynomial_ops | _coeff_map | 19 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | _build_hh_poly_and_matrix | 85 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | _build_hubbard_poly_and_matrix | 100 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | _build_hh_hamiltonian | 58 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | _run_hh_vqe | 70 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | _run_hubbard_vqe | 94 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | _run_hh_vqe_termwise | 343 |

SECTION: TEST + DOC COVERAGE | Documentation anchors (noise-scope references)

| path | module | line | snippet |
|---|---|---|---|
| README.md | test.test_adapt_vqe_integration | 115 | T0["test/test_adapt_vqe_integration.py"] |
| README.md | src.quantum.vqe_latex_python_pairs | 219 | - `src/quantum/vqe_latex_python_pairs.py` adds `expval_pauli_polynomial_c |
| README.md | src.quantum.time_propagation.cfqm_schemes | 339 | pytest -q test/test_cfqm_schemes.py test/test_cfqm_propagator.py test/test_ |
| README.md | test.test_cfqm_acceptance | 339 | pytest -q test/test_cfqm_schemes.py test/test_cfqm_propagator.py test/test_ |
| README.md | test.test_cfqm_propagator | 339 | pytest -q test/test_cfqm_schemes.py test/test_cfqm_propagator.py test/test_ |
| README.md | test.test_cfqm_schemes | 339 | pytest -q test/test_cfqm_schemes.py test/test_cfqm_propagator.py test/test_ |
| README.md | pipelines.exact_bench.hh_noise_hardware_validation | 459 | - `pipelines/exact_bench/hh_noise_hardware_validation.py` |
| README.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 465 | - `pipelines/exact_bench/hh_noise_robustness_seq_report.py` |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 125 | \| [`pipelines/exact_bench/hh_noise_hardware_validation.py`](../pipelines/ex |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 126 | \| [`pipelines/exact_bench/hh_noise_robustness_seq_report.py`](../pipelines/ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 146 | 4. JW ladder definitions should come from helper operators in [`src/quantur |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 151 | `PauliTerm` canonical source is [`src/quantum/qubitization_module.py`](../s |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 184 | Q2["src/quantum/qubitization_module.py"] |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 185 | Q3["src/quantum/pauli_polynomial_class.py"] |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 186 | Q4["src/quantum/hubbard_latex_python_pairs.py"] |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 187 | Q5["src/quantum/vqe_latex_python_pairs.py"] |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 188 | Q6["src/quantum/drives_time_potential.py"] |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 189 | Q7["src/quantum/hartree_fock_reference_state.py"] |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 246 | \| Operator algebra core \| [`src/quantum/pauli_letters_module.py`](../src/qu |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 246 | \| Operator algebra core \| [`src/quantum/pauli_letters_module.py`](../src/qu |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 247 | \| Physics builders \| [`src/quantum/hubbard_latex_python_pairs.py`](../src/qu |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 248 | \| VQE/ansatz engine \| [`src/quantum/vqe_latex_python_pairs.py`](../src/quan |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 249 | \| Drive utilities \| [`src/quantum/drives_time_potential.py`](../src/quantum/ |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 254 | \| Sequential HH robustness tooling \| [`pipelines/exact_bench/hh_noise_robus |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 254 | \| Sequential HH robustness tooling \| [`pipelines/exact_bench/hh_noise_robus |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 344 | \| New HH operator term or model component \| [`src/quantum/hubbard_latex_ |

SECTION: TEST + DOC COVERAGE | Documentation anchors (noise-scope references) (cont. 2)

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | test.test_trotter_hh_integration | 344 | \| New HH operator term or model component \| [`src/quantum/hubbard_latex_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 347 | \| Drive argument/physics extension \| [`src/quantum/drives_time_potential.p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 358 | - [`src/quantum/qubitization_module.py`](../src/quantum/qubitization_m |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 359 | - [`src/quantum/pauli_polynomial_class.py`](../src/quantum/pauli_polynom |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 368 | \| `PauliTerm` \| [`src/quantum/qubitization_module.py`](../src/quantum/qubi |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 369 | \| `PauliPolynomial` \| [`src/quantum/pauli_polynomial_class.py`](../src/quan |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 370 | \| `fermion_plus_operator` / `fermion_minus_operator` \| [`src/quantum/pauli |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 411 | Primary builder functions for model Hamiltonians live in [`src/quantum/hubb |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 430 | - [`src/quantum/vqe_latex_python_pairs.py`](../src/quantum/vqe_latex_pyth |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 526 | - [`src/quantum/drives_time_potential.py`](../src/quantum/drives_time_p |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_energy_total_observables | 553 | - test coverage in [`test/test_energy_total_observables.py`](../test/test_en |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_exact_steps_multiplier | 562 | Convergence and method-name metadata behavior is tested in [`test/test_ex |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_pauli_polynomial_ops | 689 | \| [`test/test_pauli_polynomial_ops.py`](../test/test_pauli_polynomial_ops.p |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_energy_total_observables | 691 | \| [`test/test_energy_total_observables.py`](../test/test_energy_total_obser |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_exact_steps_multiplier | 692 | \| [`test/test_exact_steps_multiplier.py`](../test/test_exact_steps_multiplie |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_time_potential_drive | 693 | \| [`test/test_time_potential_drive.py`](../test/test_time_potential_drive.py |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_trotter_hh_integration | 694 | \| [`test/test_trotter_hh_integration.py`](../test/test_trotter_hh_integration |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_vqe_hh_integration | 695 | \| [`test/test_vqe_hh_integration.py`](../test/test_vqe_hh_integration.py) \| H |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_hubbard_holstein_ansatz | 696 | \| [`test/test_hubbard_holstein_ansatz.py`](../test/test_hubbard_holstein_an |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_boson_unary | 697 | \| [`test/test_boson_unary.py`](../test/test_boson_unary.py) \| Unary boson en |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_adapt_vqe_integration | 698 | \| [`test/test_adapt_vqe_integration.py`](../test/test_adapt_vqe_integration |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_ed_crosscheck | 699 | \| [`test/test_ed_crosscheck.py`](../test/test_ed_crosscheck.py) \| ED basis/sp |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 723 | \| [`pipelines/exact_bench/hh_noise_robustness_seq_report.py`](../pipelines/ |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_exact_steps_multiplier | 731 | \| Piecewise reference method behavior and metadata \| `test/test_exact_ste |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_trotter_hh_integration | 732 | \| HH-to-Hubbard reduction sanity \| `test/test_trotter_hh_integration.py` cla |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_trotter_hh_integration | 733 | \| Compare HH guard behavior \| `test/test_trotter_hh_integration.py` class `` |

SECTION: TEST + DOC COVERAGE | Documentation anchors (noise-scope references) (cont. 3)

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | test.test_trotter_hh_integration | 734 | \| Qiskit baseline HH guard behavior \| `test/test_trotter_hh_integration.py` |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_trotter_hh_integration | 735 | \| HH exact-ground consistency helpers \| `test/test_trotter_hh_integration.p |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_adapt_vqe_integration | 736 | \| ADAPT pool behavior, including LF additions \| `test/test_adapt_vqe_integ |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_hh_seq_transition_logic | 738 | \| HH transition plateau logic \| `test/test_hh_seq_transition_logic.p |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_hh_pool_b_union | 739 | \| Pool-B strict-union dedup/provenance \| `test/test_hh_pool_b_union |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_hh_drive_qop_builder | 740 | \| Time-dependent SparsePauliOp drive merge logic \| `test/test_hh_drive_qo |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 771 | - [`pipelines/exact_bench/hh_noise_robustness_seq_report.py`](../pipelines/ |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 772 | - [`pipelines/exact_bench/hh_seq_transition_utils.py`](../pipelines/exact_be |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_hh_seq_transition_logic | 774 | - [`test/test_hh_seq_transition_logic.py`](../test/test_hh_seq_transition_ |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_hh_pool_b_union | 775 | - [`test/test_hh_pool_b_union.py`](../test/test_hh_pool_b_union.p |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_hh_drive_qop_builder | 776 | - [`test/test_hh_drive_qop_builder.py`](../test/test_hh_drive_qop_buil |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_trotter_hh_integration | 810 | - test class `TestComparePipelineHHGuard` in [`test/test_trotter_hh_integr |
| docs/LLM_RESEARCH_CONTEXT.md | test.test_trotter_hh_integration | 821 | - test class `TestQiskitBaselineHHGuard` in [`test/test_trotter_hh_integrat |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1262 | #### `pipelines/exact_bench/hh_noise_hardware_validation.p |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1265 | \| FUNC \| `_ai_log` \| `pipelines/exact_bench/hh_noise_hardware_validatio |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1266 | \| FUNC \| `_half_filled_particles` \| `pipelines/exact_bench/hh_noise_hardwar |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1267 | \| FUNC \| `_fermion_mode_index` \| `pipelines/exact_bench/hh_noise_hardwar |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1268 | \| FUNC \| `_collect_hardcoded_terms_exyz` \| `pipelines/exact_bench/hh_noise |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1269 | \| FUNC \| `_trotterized_circuit` \| `pipelines/exact_bench/hh_noise_hardware_ |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1270 | \| FUNC \| `_get_hubbard_minimum` \| `pipelines/exact_bench/hh_noise_hardwa |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1271 | \| FUNC \| `_get_hh_minimum` \| `pipelines/exact_bench/hh_noise_hardware_va |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1272 | \| FUNC \| `_apply_defaults_and_minimums` \| `pipelines/exact_bench/hh_noise |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1273 | \| FUNC \| `_build_hamiltonian` \| `pipelines/exact_bench/hh_noise_hardware_ |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1274 | \| FUNC \| `_build_ansatz` \| `pipelines/exact_bench/hh_noise_hardware_valida |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1275 | \| FUNC \| `_build_reference_state` \| `pipelines/exact_bench/hh_noise_hardwa |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1276 | \| FUNC \| `_build_adapt_pool` \| `pipelines/exact_bench/hh_noise_hardware_v |

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1277 | \| FUNC \| `_adapt_ops_to_circuit` \| `pipelines/exact_bench/hh_noise_hardwar |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1278 | \| FUNC \| `_run_noisy_adapt` \| `pipelines/exact_bench/hh_noise_hardware_va |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1279 | \| FUNC \| `_run_noisy_vqe` \| `pipelines/exact_bench/hh_noise_hardware_valid |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1280 | \| FUNC \| `_run_noisy_trotter` \| `pipelines/exact_bench/hh_noise_hardware_v |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1281 | \| FUNC \| `_initial_state_selection` \| `pipelines/exact_bench/hh_noise_hardw |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1282 | \| FUNC \| `_parse_compare_observables` \| `pipelines/exact_bench/hh_noise_h |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1283 | \| FUNC \| `_load_legacy_trajectory` \| `pipelines/exact_bench/hh_noise_hardw |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1284 | \| FUNC \| `_compute_legacy_parity` \| `pipelines/exact_bench/hh_noise_hardw |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1285 | \| FUNC \| `_write_legacy_comparison_plot` \| `pipelines/exact_bench/hh_noise |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1286 | \| FUNC \| `_write_noise_validation_pdf` \| `pipelines/exact_bench/hh_noise_ha |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1287 | \| FUNC \| `parse_args` \| `pipelines/exact_bench/hh_noise_hardware_validati |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_hardware_validation | 1288 | \| FUNC \| `main` \| `pipelines/exact_bench/hh_noise_hardware_validation |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1290 | #### `pipelines/exact_bench/hh_noise_robustness_seq_report. |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1293 | \| FUNC \| `_ai_log` \| `pipelines/exact_bench/hh_noise_robustness_seq_rep |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1294 | \| FUNC \| `_half_filled_particles` \| `pipelines/exact_bench/hh_noise_robustne |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1295 | \| FUNC \| `_collect_hardcoded_terms_exyz` \| `pipelines/exact_bench/hh_noise |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1296 | \| FUNC \| `_normalize_state` \| `pipelines/exact_bench/hh_noise_robustness_s |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1297 | \| FUNC \| `_build_hh_hamiltonian` \| `pipelines/exact_bench/hh_noise_robustn |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1298 | \| FUNC \| `_build_reference_state` \| `pipelines/exact_bench/hh_noise_robustn |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1299 | \| FUNC \| `_build_hh_hva_ansatz` \| `pipelines/exact_bench/hh_noise_robustne |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1300 | \| FUNC \| `_apply_pauli_polynomial` \| `pipelines/exact_bench/hh_noise_robus |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1301 | \| FUNC \| `_prepare_state` \| `pipelines/exact_bench/hh_noise_robustness_seq |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1302 | \| FUNC \| `_energy` \| `pipelines/exact_bench/hh_noise_robustness_seq_rep |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1303 | \| FUNC \| `_commutator_gradient` \| `pipelines/exact_bench/hh_noise_robustn |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1304 | \| FUNC \| `_build_uccsd_fermion_lifted_pool` \| `pipelines/exact_bench/hh_noi |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1305 | \| FUNC \| `_build_hva_pool` \| `pipelines/exact_bench/hh_noise_robustness_se |

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1306 | \| FUNC \| `_build_paop_full_pool` \| `pipelines/exact_bench/hh_noise_robustne |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1307 | \| FUNC \| `_run_vqe_stage_with_transition` \| `pipelines/exact_bench/hh_nois |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1308 | \| FUNC \| `_run_adapt_stage_with_transition` \| `pipelines/exact_bench/hh_no |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1309 | \| FUNC \| `_parse_custom_s` \| `pipelines/exact_bench/hh_noise_robustness_se |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1310 | \| FUNC \| `_build_drive_profile` \| `pipelines/exact_bench/hh_noise_robustnes |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1311 | \| FUNC \| `_drive_provider_from_profile` \| `pipelines/exact_bench/hh_noise_r |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1312 | \| FUNC \| `_spin_orbital_bit_index` \| `pipelines/exact_bench/hh_noise_robust |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1313 | \| FUNC \| `_staggered_qop` \| `pipelines/exact_bench/hh_noise_robustness_se |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1314 | \| FUNC \| `_time_sample` \| `pipelines/exact_bench/hh_noise_robustness_seq_ |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1315 | \| FUNC \| `_build_suzuki2_time_dependent_circuit` \| `pipelines/exact_bench/h |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1316 | \| FUNC \| `_run_noisy_suzuki_trajectory` \| `pipelines/exact_bench/hh_noise_r |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1317 | \| FUNC \| `_noisy_worker_entry` \| `pipelines/exact_bench/hh_noise_robustnes |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1318 | \| FUNC \| `_run_noisy_mode_isolated` \| `pipelines/exact_bench/hh_noise_robu |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1319 | \| FUNC \| `_extract_series` \| `pipelines/exact_bench/hh_noise_robustness_se |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1320 | \| FUNC \| `_compute_exact_reference_for_hh` \| `pipelines/exact_bench/hh_no |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1321 | \| FUNC \| `_run_hardcoded_suzuki_profile` \| `pipelines/exact_bench/hh_noise_ |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1322 | \| FUNC \| `_run_noiseless_profile` \| `pipelines/exact_bench/hh_noise_robustn |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1323 | \| FUNC \| `_compute_comparisons` \| `pipelines/exact_bench/hh_noise_robustn |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1324 | \| FUNC \| `_build_summary` \| `pipelines/exact_bench/hh_noise_robustness_se |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1325 | \| FUNC \| `_build_equation_registry_and_contracts` \| `pipelines/exact_bench/ |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1326 | \| FUNC \| `_apply_report_theme` \| `pipelines/exact_bench/hh_noise_robustnes |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1327 | \| FUNC \| `_render_page_header_footer` \| `pipelines/exact_bench/hh_noise_ro |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1328 | \| FUNC \| `_build_caption_overrides` \| `pipelines/exact_bench/hh_noise_robus |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1329 | \| FUNC \| `_fallback_caption_line` \| `pipelines/exact_bench/hh_noise_robustn |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1330 | \| FUNC \| `_extract_numeric_caption_line` \| `pipelines/exact_bench/hh_noise_ |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1331 | \| FUNC \| `_latex_to_unicode_math` \| `pipelines/exact_bench/hh_noise_robust |

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1332 | \| FUNC \| `_latex_to_mathtext` \| `pipelines/exact_bench/hh_noise_robustness |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1333 | \| FUNC \| `_annotate_plot_with_equations` \| `pipelines/exact_bench/hh_noise |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1334 | \| FUNC \| `_matrix_from_rows` \| `pipelines/exact_bench/hh_noise_robustness |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1335 | \| FUNC \| `_render_formula_atlas` \| `pipelines/exact_bench/hh_noise_robustn |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1336 | \| FUNC \| `_write_pdf` \| `pipelines/exact_bench/hh_noise_robustness_seq_rep |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1337 | \| FUNC \| `_json_dump` \| `pipelines/exact_bench/hh_noise_robustness_seq_re |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1338 | \| FUNC \| `_enforce_defaults_and_minimums` \| `pipelines/exact_bench/hh_noi |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1339 | \| FUNC \| `parse_args` \| `pipelines/exact_bench/hh_noise_robustness_seq_re |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 1340 | \| FUNC \| `main` \| `pipelines/exact_bench/hh_noise_robustness_seq_repor |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1342 | #### `pipelines/exact_bench/hh_seq_transition_utils.py` |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1345 | \| FUNC \| `_to_ixyz` \| `pipelines/exact_bench/hh_seq_transition_utils.p |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1346 | \| FUNC \| `polynomial_signature` \| `pipelines/exact_bench/hh_seq_transition |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1347 | \| CLASS \| `TransitionConfig` \| `pipelines/exact_bench/hh_seq_transition_u |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1348 | \| CLASS \| `TransitionState` \| `pipelines/exact_bench/hh_seq_transition_u |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1349 | \| FUNC \| `_window_slope` \| `pipelines/exact_bench/hh_seq_transition_ut |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1350 | \| FUNC \| `update_transition_state` \| `pipelines/exact_bench/hh_seq_transit |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1351 | \| FUNC \| `summarize_transition` \| `pipelines/exact_bench/hh_seq_transition |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1352 | \| FUNC \| `build_pool_b_strict_union` \| `pipelines/exact_bench/hh_seq_trans |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1353 | \| FUNC \| `build_time_dependent_sparse_qop` \| `pipelines/exact_bench/hh_se |
| docs/LLM_RESEARCH_CONTEXT.md | pipelines.exact_bench.hh_seq_transition_utils | 1354 | \| FUNC \| `flatten_coeff_map_real_imag` \| `pipelines/exact_bench/hh_seq_tr |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1356 | #### `src/quantum/drives_time_potential.py` |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1359 | \| FUNC \| `reference_method_name` \| `src/quantum/drives_time_potenti |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1360 | \| FUNC \| `gaussian_sinusoid_waveform` \| `src/quantum/drives_time_poten |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1361 | \| FUNC \| `evaluate_drive_waveform` \| `src/quantum/drives_time_potenti |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1362 | \| FUNC \| `default_spatial_weights` \| `src/quantum/drives_time_potentia |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1363 | \| CLASS \| `GaussianSinusoidSitePotential` \| `src/quantum/drives_time_pote |

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1364 | \| METHOD \| `GaussianSinusoidSitePotential.v_scalar` \| `src/quantum/drives_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1365 | \| METHOD \| `GaussianSinusoidSitePotential.v_sites` \| `src/quantum/drives_t |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1366 | \| FUNC \| `_z_label_exyz` \| `src/quantum/drives_time_potential.py:2 |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1367 | \| CLASS \| `DensityDriveTemplate` \| `src/quantum/drives_time_potential |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1368 | \| METHOD \| `DensityDriveTemplate.build` \| `src/quantum/drives_time_pote |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1369 | \| METHOD \| `DensityDriveTemplate.labels_exyz` \| `src/quantum/drives_time_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1370 | \| METHOD \| `DensityDriveTemplate.z_label` \| `src/quantum/drives_time_pote |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1371 | \| CLASS \| `TimeDependentOnsiteDensityDrive` \| `src/quantum/drives_time_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1372 | \| METHOD \| `TimeDependentOnsiteDensityDrive.coeff_map_exyz` \| `src/quant |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.drives_time_potential | 1373 | \| FUNC \| `build_gaussian_sinusoid_density_drive` \| `src/quantum/drives_tim |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 1375 | #### `src/quantum/hartree_fock_reference_state.py` |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 1378 | \| FUNC \| `_validate_indexing` \| `src/quantum/hartree_fock_reference_sta |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 1379 | \| FUNC \| `mode_index` \| `src/quantum/hartree_fock_reference_state. |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 1380 | \| FUNC \| `hartree_fock_occupied_qubits` \| `src/quantum/hartree_fock_refere |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 1381 | \| FUNC \| `bitstring_qn1_to_q0` \| `src/quantum/hartree_fock_reference_sta |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 1382 | \| FUNC \| `hartree_fock_bitstring` \| `src/quantum/hartree_fock_reference_st |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 1383 | \| FUNC \| `hartree_fock_statevector` \| `src/quantum/hartree_fock_reference_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 1384 | \| FUNC \| `_half_filled_num_particles` \| `src/quantum/hartree_fock_reference |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 1385 | \| FUNC \| `_phonon_vacuum_bitstring` \| `src/quantum/hartree_fock_reference |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hartree_fock_reference_state | 1386 | \| FUNC \| `hubbard_holstein_reference_state` \| `src/quantum/hartree_fock_re |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1388 | #### `src/quantum/hubbard_latex_python_pairs.py` |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1391 | \| CLASS \| `_FallbackLog` \| `src/quantum/hubbard_latex_python_pairs. |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1392 | \| METHOD \| `_FallbackLog.error` \| `src/quantum/hubbard_latex_python_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1393 | \| FUNC \| `mode_index` \| `src/quantum/hubbard_latex_python_pairs.p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1394 | \| FUNC \| `_prod` \| `src/quantum/hubbard_latex_python_pairs.py:1 |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1395 | \| FUNC \| `n_sites_from_dims` \| `src/quantum/hubbard_latex_python_pair |

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1396 | \| FUNC \| `coord_to_site_index` \| `src/quantum/hubbard_latex_python_pai |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1397 | \| FUNC \| `bravais_nearest_neighbor_edges` \| `src/quantum/hubbard_latex_py |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1398 | \| FUNC \| `jw_number_operator` \| `src/quantum/hubbard_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1399 | \| FUNC \| `_resolve_hubbard_nq` \| `src/quantum/hubbard_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1400 | \| FUNC \| `boson_qubits_per_site` \| `src/quantum/hubbard_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1401 | \| FUNC \| `phonon_qubit_indices_for_site` \| `src/quantum/hubbard_latex_pyth |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1402 | \| FUNC \| `_normalize_pauli_symbol` \| `src/quantum/hubbard_latex_python_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1403 | \| FUNC \| `_identity_pauli_polynomial` \| `src/quantum/hubbard_latex_python_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1404 | \| FUNC \| `_pauli_monomial_global` \| `src/quantum/hubbard_latex_python_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1405 | \| FUNC \| `_boson_local_matrices` \| `src/quantum/hubbard_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1406 | \| FUNC \| `_embed_to_qubit_space` \| `src/quantum/hubbard_latex_python_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1407 | \| FUNC \| `_pauli_basis_mats` \| `src/quantum/hubbard_latex_python_pair |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1408 | \| FUNC \| `_matrix_to_pauli_coeffs` \| `src/quantum/hubbard_latex_python_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1409 | \| FUNC \| `boson_local_operator_pauli_decomp` \| `src/quantum/hubbard_latex |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1410 | \| FUNC \| `boson_operator` \| `src/quantum/hubbard_latex_python_pairs |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1411 | \| FUNC \| `boson_number_operator` \| `src/quantum/hubbard_latex_python_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1412 | \| FUNC \| `boson_displacement_operator` \| `src/quantum/hubbard_latex_pytho |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1413 | \| FUNC \| `_unary_site_qubits_check` \| `src/quantum/hubbard_latex_python_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1414 | \| FUNC \| `boson_unary_number_operator` \| `src/quantum/hubbard_latex_pyth |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1415 | \| FUNC \| `boson_unary_displacement_operator` \| `src/quantum/hubbard_latex |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1416 | \| FUNC \| `boson_unary_bdag_operator` \| `src/quantum/hubbard_latex_pyth |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1417 | \| FUNC \| `boson_unary_b_operator` \| `src/quantum/hubbard_latex_python_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1418 | \| FUNC \| `build_hubbard_kinetic` \| `src/quantum/hubbard_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1419 | \| FUNC \| `build_hubbard_onsite` \| `src/quantum/hubbard_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1420 | \| FUNC \| `_parse_site_potential` \| `src/quantum/hubbard_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1421 | \| FUNC \| `build_hubbard_potential` \| `src/quantum/hubbard_latex_python_p |

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1422 | \| FUNC \| `build_hubbard_hamiltonian` \| `src/quantum/hubbard_latex_python_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1423 | \| FUNC \| `_eval_site_potential_maybe_time_dependent` \| `src/quantum/hubba |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1424 | \| FUNC \| `build_holstein_phonon_energy` \| `src/quantum/hubbard_latex_pyth |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1425 | \| FUNC \| `build_holstein_coupling` \| `src/quantum/hubbard_latex_python_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1426 | \| FUNC \| `build_hubbard_holstein_drive` \| `src/quantum/hubbard_latex_pytho |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1427 | \| FUNC \| `build_hubbard_holstein_hamiltonian` \| `src/quantum/hubbard_latex |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1428 | \| FUNC \| `show_latex_and_code` \| `src/quantum/hubbard_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.hubbard_latex_python_pairs | 1429 | \| FUNC \| `show_hubbard_latex_python_pairs` \| `src/quantum/hubbard_latex_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1431 | #### `src/quantum/vqe_latex_python_pairs.py` |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1434 | \| CLASS \| `_FallbackLog` \| `src/quantum/vqe_latex_python_pairs.py |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1435 | \| METHOD \| `_FallbackLog.error` \| `src/quantum/vqe_latex_python_pair |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1436 | \| METHOD \| `_FallbackLog.info` \| `src/quantum/vqe_latex_python_pairs |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1437 | \| FUNC \| `_normalize_pauli_string` \| `src/quantum/vqe_latex_python_pai |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1438 | \| FUNC \| `basis_state` \| `src/quantum/vqe_latex_python_pairs.py:2 |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1439 | \| FUNC \| `apply_pauli_string` \| `src/quantum/vqe_latex_python_pairs.p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1440 | \| FUNC \| `expval_pauli_string` \| `src/quantum/vqe_latex_python_pairs. |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1441 | \| FUNC \| `expval_pauli_polynomial` \| `src/quantum/vqe_latex_python_pai |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1442 | \| FUNC \| `apply_pauli_rotation` \| `src/quantum/vqe_latex_python_pairs |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1443 | \| FUNC \| `apply_exp_pauli_polynomial` \| `src/quantum/vqe_latex_python_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1444 | \| FUNC \| `half_filled_num_particles` \| `src/quantum/vqe_latex_python_pai |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1445 | \| FUNC \| `jw_number_operator` \| `src/quantum/vqe_latex_python_pairs |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1446 | \| FUNC \| `hubbard_hop_term` \| `src/quantum/vqe_latex_python_pairs. |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1447 | \| FUNC \| `hubbard_onsite_term` \| `src/quantum/vqe_latex_python_pairs |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1448 | \| FUNC \| `hubbard_potential_term` \| `src/quantum/vqe_latex_python_pai |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1449 | \| FUNC \| `_parse_site_potential` \| `src/quantum/vqe_latex_python_pairs |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1450 | \| FUNC \| `_single_term_polynomials_sorted` \| `src/quantum/vqe_latex_pytho |

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1451 | \| CLASS \| `AnsatzTerm` \| `src/quantum/vqe_latex_python_pairs.py:4 |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1452 | \| CLASS \| `HubbardTermwiseAnsatz` \| `src/quantum/vqe_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1453 | \| METHOD \| `HubbardTermwiseAnsatz.__init__` \| `src/quantum/vqe_latex_pyth |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1454 | \| METHOD \| `HubbardTermwiseAnsatz._build_base_terms` \| `src/quantum/vqe_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1455 | \| METHOD \| `HubbardTermwiseAnsatz.prepare_state` \| `src/quantum/vqe_late |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1456 | \| CLASS \| `HubbardLayerwiseAnsatz` \| `src/quantum/vqe_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1457 | \| METHOD \| `HubbardLayerwiseAnsatz._build_base_terms` \| `src/quantum/vqe_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1458 | \| METHOD \| `HubbardLayerwiseAnsatz.prepare_state` \| `src/quantum/vqe_late |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1459 | \| CLASS \| `HardcodedUCCSDAnsatz` \| `src/quantum/vqe_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1460 | \| METHOD \| `HardcodedUCCSDAnsatz.__init__` \| `src/quantum/vqe_latex_pyth |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1461 | \| METHOD \| `HardcodedUCCSDAnsatz._single_generator` \| `src/quantum/vqe_l |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1462 | \| METHOD \| `HardcodedUCCSDAnsatz._double_generator` \| `src/quantum/vqe_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1463 | \| METHOD \| `HardcodedUCCSDAnsatz._build_base_terms` \| `src/quantum/vqe_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1464 | \| METHOD \| `HardcodedUCCSDAnsatz.prepare_state` \| `src/quantum/vqe_latex |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1465 | \| CLASS \| `HardcodedUCCSDLayerwiseAnsatz` \| `src/quantum/vqe_latex_pytho |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1466 | \| METHOD \| `HardcodedUCCSDLayerwiseAnsatz._build_base_terms` \| `src/quan |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1467 | \| METHOD \| `HardcodedUCCSDLayerwiseAnsatz.prepare_state` \| `src/quantum/ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1468 | \| CLASS \| `HubbardHolsteinTermwiseAnsatz` \| `src/quantum/vqe_latex_pytho |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1469 | \| METHOD \| `HubbardHolsteinTermwiseAnsatz.__init__` \| `src/quantum/vqe_la |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1470 | \| METHOD \| `HubbardHolsteinTermwiseAnsatz._build_base_terms` \| `src/quant |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1471 | \| METHOD \| `HubbardHolsteinTermwiseAnsatz.prepare_state` \| `src/quantum/v |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1472 | \| CLASS \| `HubbardHolsteinPhysicalTermwiseAnsatz` \| `src/quantum/vqe_late |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1473 | \| METHOD \| `HubbardHolsteinPhysicalTermwiseAnsatz.__init__` \| `src/quantu |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1474 | \| METHOD \| `HubbardHolsteinPhysicalTermwiseAnsatz._build_base_terms` \| `s |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1475 | \| METHOD \| `HubbardHolsteinPhysicalTermwiseAnsatz.prepare_state` \| `src/q |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1476 | \| CLASS \| `HubbardHolsteinLayerwiseAnsatz` \| `src/quantum/vqe_latex_pytho |

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1477 | \| METHOD \| `HubbardHolsteinLayerwiseAnsatz.__init__` \| `src/quantum/vqe_la |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1478 | \| METHOD \| `HubbardHolsteinLayerwiseAnsatz._poly_group` \| `src/quantum/vq |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1479 | \| METHOD \| `HubbardHolsteinLayerwiseAnsatz._build_base_terms` \| `src/quan |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1480 | \| METHOD \| `HubbardHolsteinLayerwiseAnsatz.prepare_state` \| `src/quantum/ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1481 | \| CLASS \| `VQEResult` \| `src/quantum/vqe_latex_python_pairs.py:1 |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1482 | \| FUNC \| `_try_import_scipy_minimize` \| `src/quantum/vqe_latex_python_pa |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1483 | \| FUNC \| `vqe_minimize` \| `src/quantum/vqe_latex_python_pairs.py: |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1484 | \| FUNC \| `pauli_matrix` \| `src/quantum/vqe_latex_python_pairs.py:1 |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1485 | \| FUNC \| `hamiltonian_matrix` \| `src/quantum/vqe_latex_python_pairs.p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1486 | \| FUNC \| `_spin_orbital_index_sets` \| `src/quantum/vqe_latex_python_pair |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1487 | \| FUNC \| `_sector_basis_indices` \| `src/quantum/vqe_latex_python_pairs |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1488 | \| FUNC \| `exact_ground_energy_sector` \| `src/quantum/vqe_latex_python_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1489 | \| FUNC \| `_sector_basis_indices_fermion_only` \| `src/quantum/vqe_latex_pyt |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1490 | \| FUNC \| `exact_ground_energy_sector_hh` \| `src/quantum/vqe_latex_python_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1491 | \| FUNC \| `show_latex_and_code` \| `src/quantum/vqe_latex_python_pairs |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.vqe_latex_python_pairs | 1492 | \| FUNC \| `show_vqe_latex_python_pairs` \| `src/quantum/vqe_latex_python_p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1515 | #### `src/quantum/pauli_polynomial_class.py` |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1518 | \| CLASS \| `PauliPolynomial` \| `src/quantum/pauli_polynomial_class.p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1519 | \| METHOD \| `PauliPolynomial.__init__` \| `src/quantum/pauli_polynomial_c |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1520 | \| METHOD \| `PauliPolynomial.get_nq` \| `src/quantum/pauli_polynomial_cl |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1521 | \| METHOD \| `PauliPolynomial.return_polynomial` \| `src/quantum/pauli_polyn |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1522 | \| METHOD \| `PauliPolynomial.count_number_terms` \| `src/quantum/pauli_poly |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1523 | \| METHOD \| `PauliPolynomial.add_term` \| `src/quantum/pauli_polynomial_c |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1524 | \| METHOD \| `PauliPolynomial._clone_term` \| `src/quantum/pauli_polynomial |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1525 | \| METHOD \| `PauliPolynomial._clone_terms` \| `src/quantum/pauli_polynomia |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1526 | \| METHOD \| `PauliPolynomial.__add__` \| `src/quantum/pauli_polynomial_c |

| path | module | line | snippet |
|---|---|---|---|
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1527 | \| METHOD \| `PauliPolynomial.__iadd__` \| `src/quantum/pauli_polynomial_c |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1528 | \| METHOD \| `PauliPolynomial.__sub__` \| `src/quantum/pauli_polynomial_c |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1529 | \| METHOD \| `PauliPolynomial.__isub__` \| `src/quantum/pauli_polynomial_c |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1530 | \| METHOD \| `PauliPolynomial.__mul__` \| `src/quantum/pauli_polynomial_c |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1531 | \| METHOD \| `PauliPolynomial.__rmul__` \| `src/quantum/pauli_polynomial_cl |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1532 | \| METHOD \| `PauliPolynomial.__pow__` \| `src/quantum/pauli_polynomial_cl |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1533 | \| METHOD \| `PauliPolynomial._reduce` \| `src/quantum/pauli_polynomial_cl |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1534 | \| METHOD \| `PauliPolynomial.visualize_polynomial` \| `src/quantum/pauli_pol |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1535 | \| CLASS \| `fermion_plus_operator` \| `src/quantum/pauli_polynomial_clas |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1536 | \| METHOD \| `fermion_plus_operator.__init__` \| `src/quantum/pauli_polynomia |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1537 | \| METHOD \| `fermion_plus_operator.__set_JW_operator` \| `src/quantum/pauli_ |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1538 | \| CLASS \| `fermion_minus_operator` \| `src/quantum/pauli_polynomial_cla |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1539 | \| METHOD \| `fermion_minus_operator.__init__` \| `src/quantum/pauli_polynom |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.pauli_polynomial_class | 1540 | \| METHOD \| `fermion_minus_operator.__set_JW_operator` \| `src/quantum/pau |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 1545 | #### `src/quantum/qubitization_module.py` |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 1548 | \| CLASS \| `PauliTerm` \| `src/quantum/qubitization_module.py:10 |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 1549 | \| METHOD \| `PauliTerm.__init__` \| `src/quantum/qubitization_module. |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 1550 | \| METHOD \| `PauliTerm.__set_pauli_term` \| `src/quantum/qubitization_mod |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 1551 | \| METHOD \| `PauliTerm.__set_pauli_term_from_letters` \| `src/quantum/qubiti |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 1552 | \| METHOD \| `PauliTerm.nqubit` \| `src/quantum/qubitization_module.p |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 1553 | \| METHOD \| `PauliTerm.__mul__` \| `src/quantum/qubitization_module. |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 1554 | \| METHOD \| `PauliTerm.pw2strng` \| `src/quantum/qubitization_module |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 1555 | \| METHOD \| `PauliTerm.pw2sparsePauliOp` \| `src/quantum/qubitization_mo |
| docs/LLM_RESEARCH_CONTEXT.md | src.quantum.qubitization_module | 1556 | \| METHOD \| `PauliTerm.visualize` \| `src/quantum/qubitization_module |
| docs/repo_implementation_guide.md | src.quantum.qubitization_module | 72 | Canonical class is `src/quantum/qubitization_module.py` with compatibility |
| docs/repo_implementation_guide.md | src.quantum.hubbard_latex_python_pairs | 108 | - `src/quantum/hubbard_latex_python_pairs.py` |

| path | module | line | snippet |
|---|---|---|---|
| docs/repo_implementation_guide.md | src.quantum.vqe_latex_python_pairs | 109 | - `src/quantum/vqe_latex_python_pairs.py` |
| docs/repo_implementation_guide.md | src.quantum.hartree_fock_reference_state | 110 | - `src/quantum/hartree_fock_reference_state.py` |
| docs/repo_implementation_guide.md | src.quantum.drives_time_potential | 111 | - `src/quantum/drives_time_potential.py` |
| docs/repo_implementation_guide.md | src.quantum.time_propagation.cfqm_schemes | 113 | - `src/quantum/time_propagation/cfqm_schemes.py` |
| docs/repo_implementation_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 119 | - `pipelines/exact_bench/hh_noise_hardware_validation.py` |
| docs/repo_implementation_guide.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 120 | - `pipelines/exact_bench/hh_noise_robustness_seq_report.py` |
| docs/repo_implementation_guide.md | pipelines.exact_bench.hh_seq_transition_utils | 121 | - `pipelines/exact_bench/hh_seq_transition_utils.py` |
| docs/repo_implementation_guide.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 528 | - `pipelines/exact_bench/hh_noise_robustness_seq_report.py` |
| docs/repo_implementation_guide.md | pipelines.exact_bench.hh_seq_transition_utils | 529 | - `pipelines/exact_bench/hh_seq_transition_utils.py`, |
| docs/repo_implementation_guide.md | test.test_hh_seq_transition_logic | 532 | - `test/test_hh_seq_transition_logic.py`, |
| docs/repo_implementation_guide.md | test.test_hh_pool_b_union | 533 | - `test/test_hh_pool_b_union.py`, |
| docs/repo_implementation_guide.md | test.test_hh_drive_qop_builder | 534 | - `test/test_hh_drive_qop_builder.py`. |
| pipelines/exact_bench/README.md | test.test_ed_crosscheck | 26 | - `test/test_ed_crosscheck.py` → "does the ED module compute correct ei |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_robustness_seq_report | 525 | - `pipelines/exact_bench/hh_noise_robustness_seq_report.py` |
| pipelines/run_guide.md | src.quantum.vqe_latex_python_pairs | 750 | - Legacy term-wise classes remain available in `src/quantum/vqe_latex_pyth |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1412 | Use `pipelines/exact_bench/hh_noise_hardware_validation.py` to validate no |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1417 | python pipelines/exact_bench/hh_noise_hardware_validation.py |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1426 | python pipelines/exact_bench/hh_noise_hardware_validation.py |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1435 | python pipelines/exact_bench/hh_noise_hardware_validation.py |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1445 | python pipelines/exact_bench/hh_noise_hardware_validation.py |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1497 | `pipelines/exact_bench/hh_noise_hardware_validation.py` now supports an o |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1502 | python pipelines/exact_bench/hh_noise_hardware_validation.py |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1514 | python pipelines/exact_bench/hh_noise_hardware_validation.py |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1539 | `hh_noise_hardware_validation.py` now supports reliability flags for constr |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1569 | New CLI flags in `hh_noise_hardware_validation.py`: |
| pipelines/run_guide.md | pipelines.exact_bench.hh_noise_hardware_validation | 1588 | python pipelines/exact_bench/hh_noise_hardware_validation.py |

```
SECTION: REFACTOR PLAYBOOK

Safe-change boundaries:
  - Keep operator-core files untouched (pauli_letters_module.py, pauli_words.py, qubitization_module.py).
  - Keep Pauli ordering and exyz/ixyz boundary conversion conventions intact.
  - Keep legacy parity gate semantics (exact time grid + strict tolerance).
  - Keep AER fallback semantics opt-in/opt-out via flags.

When extending noise behavior:
  1. update CLI parse + help text
  2. update JSON schema docs/run-guide sections
  3. add/adjust unit tests for new contract branches
  4. preserve manifest-first PDF generation

Regression checklist:
  - test_hh_noise_model_repo_guide.py
  - test_hh_noise_validation_cli.py
  - test_hh_noise_oracle_runtime.py
  - test_hh_noise_robustness_benchmarks.py
```

SECTION: APPENDIX SYMBOL INDEX | Full symbol index for resolved noise scope

| path | module | kind | name | line |
|---|---|---|---|---|
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | CONST | REPO_ROOT | 25 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _ai_log | 78 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | CONST | _HUBBARD_PARAMS | 83 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | CONST | _HH_PARAMS | 91 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _half_filled_particles | 98 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _fermion_mode_index | 102 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _collect_hardcoded_terms_exyz | 114 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | CONST | _SUZUKI2_MATH | 131 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _trotterized_circuit | 134 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _get_hubbard_minimum | 154 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _get_hh_minimum | 166 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _apply_defaults_and_minimums | 180 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _build_hamiltonian | 232 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _build_ansatz | 261 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _build_reference_state | 344 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _build_adapt_pool | 370 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _adapt_ops_to_circuit | 448 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _run_noisy_adapt | 472 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _run_noisy_vqe | 669 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _run_noisy_trotter | 780 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _initial_state_selection | 822 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _parse_compare_observables | 874 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _load_legacy_trajectory | 897 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _compute_legacy_parity | 928 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _write_legacy_comparison_plot | 987 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | _write_noise_validation_pdf | 1034 |

| path | module | kind | name | line |
|---|---|---|---|---|
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | parse_args | 1175 |
| elines/exact_bench/hh_noise_hardware_valida | pipelines.exact_bench.hh_noise_hardware_valida | FUNC | main | 1262 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | CONST | REPO_ROOT | 30 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _ai_log | 98 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | CONST | _HH_MINIMUMS | 107 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | CONST | _NOISY_METHODS_ALLOWED | 113 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _half_filled_particles | 116 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _collect_hardcoded_terms_exyz | 120 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _parse_noisy_methods_csv | 137 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _normalize_state | 154 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_hh_hamiltonian | 162 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_reference_state | 181 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_hh_hva_ansatz | 196 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _apply_pauli_polynomial | 212 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _prepare_state | 231 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _energy | 238 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _commutator_gradient | 243 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_uccsd_fermion_lifted_pool | 250 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_hva_pool | 293 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_paop_full_pool | 297 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _run_vqe_stage_with_transition | 318 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _run_adapt_stage_with_transition | 414 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _parse_custom_s | 551 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_drive_profile | 567 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _drive_provider_from_profile | 587 |
| lines/exact_bench/hh_noise_robustness_seq_r | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _spin_orbital_bit_index | 616 |

| path | module | kind | name | line |
|---|---|---|---|---|
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _staggered_qop | 625 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _time_sample | 637 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_suzuki2_time_dependent_circuit | 648 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_cfqm_stage_map_exyz | 683 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_cfqm_time_dependent_circuit | 721 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _pauli_weight | 772 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _pauli_xy_count | 776 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _cx_proxy_term | 780 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _sq_proxy_term | 784 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _active_labels_exyz | 788 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _compute_sweep_proxy_cost | 801 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _compute_time_dynamics_proxy_cost | 812 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _run_noisy_suzuki_trajectory | 888 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _run_noisy_method_trajectory | 1054 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _noisy_worker_entry | 1275 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _run_noisy_mode_isolated | 1283 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _extract_series | 1333 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _compute_exact_reference_for_hh | 1337 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _run_hardcoded_suzuki_profile | 1363 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _run_noiseless_profile | 1439 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _compute_comparisons | 1539 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_summary | 1631 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_equation_registry_and_contracts | 1675 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _apply_report_theme | 1991 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _render_page_header_footer | 2010 |
| lines/exact_bench/hh_noise_robustness_seq_ | pipelines.exact_bench.hh_noise_robustness_seq_ | CONST | _ACTIVE_CAPTION_OVERRIDES | 2015 |

| path | module | kind | name | line |
|---|---|---|---|---|
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _build_caption_overrides | 2018 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _fallback_caption_line | 2143 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _extract_numeric_caption_line | 2162 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _latex_to_unicode_math | 2170 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _latex_to_mathtext | 2212 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _annotate_plot_with_equations | 2225 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _matrix_from_rows | 2272 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _render_formula_atlas | 2276 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _write_pdf | 2344 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _json_dump | 3378 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _enforce_defaults_and_minimums | 3383 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | parse_args | 3435 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _validate_pool_b_strict_composition | 3532 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | _collect_noisy_benchmark_rows | 3566 |
| lines/exact_bench/hh_noise_robustness_spiral | pipelines.exact_bench.hh_noise_robustness_seq_ | FUNC | main | 3600 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | FUNC | _to_ixyz | 19 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | FUNC | polynomial_signature | 29 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | CLASS | TransitionConfig | 46 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | CLASS | TransitionState | 56 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | FUNC | _window_slope | 68 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | FUNC | update_transition_state | 83 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | FUNC | summarize_transition | 133 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | FUNC | build_pool_b_strict_union | 152 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | FUNC | build_time_dependent_sparse_qop | 234 |
| pipelines/exact_bench/hh_seq_transition_utils | pipelines.exact_bench.hh_seq_transition_util | FUNC | flatten_coeff_map_real_imag | 263 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | CLASS | OracleConfig | 27 |

| path | module | kind | name | line |
|---|---|---|---|---|
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | CLASS | OracleEstimate | 43 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | CLASS | NoiseBackendInfo | 52 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _to_ixyz | 60 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | CONST | _PAULI_POLY_TO_QOP_MATH | 70 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _pauli_poly_to_sparse_pauli_op | 73 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _ansatz_terms_with_parameters | 94 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _append_reference_state | 130 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _ansatz_to_circuit | 156 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _load_fake_backend | 182 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _resolve_noise_backend | 203 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | CONST | _OMP_SHM_MARKERS | 231 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | CONST | _AER_PREFLIGHT_OK_CACHE | 237 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _tail_text | 240 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _looks_like_openmp_shm_abort | 249 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _apply_omp_env_workaround | 256 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _preflight_aer_environment | 269 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _build_estimator | 349 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _extract_expectation_value | 467 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _run_estimator_job | 501 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _term_measurement_circuit | 528 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _pauli_expectation_from_counts | 551 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _run_sampler_fallback_job | 575 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | CLASS | ExpectationOracle | 593 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | CONST | _NUMBER_OPERATOR_MATH | 736 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _number_operator_qop | 739 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | CONST | _DOUBLON_OPERATOR_MATH | 753 |

| path | module | kind | name | line |
|---|---|---|---|---|
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _doublon_site_qop | 756 |
| pipelines/exact_bench/noise_oracle_runtime.p | pipelines.exact_bench.noise_oracle_runtime | FUNC | _ordered_qop_from_exyz | 776 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | CONST | REFERENCE_METHOD_NAMES | 75 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | FUNC | reference_method_name | 82 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | FUNC | gaussian_sinusoid_waveform | 119 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | FUNC | evaluate_drive_waveform | 137 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | FUNC | default_spatial_weights | 177 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | CLASS | GaussianSinusoidSitePotential | 226 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | FUNC | _z_label_exyz | 255 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | CLASS | DensityDriveTemplate | 268 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | CLASS | TimeDependentOnsiteDensityDrive | 334 |
| src/quantum/drives_time_potential.py | src.quantum.drives_time_potential | FUNC | build_gaussian_sinusoid_density_drive | 385 |
| src/quantum/hartree_fock_reference_state.p | src.quantum.hartree_fock_reference_state | FUNC | _validate_indexing | 48 |
| src/quantum/hartree_fock_reference_state.p | src.quantum.hartree_fock_reference_state | FUNC | mode_index | 57 |
| src/quantum/hartree_fock_reference_state.p | src.quantum.hartree_fock_reference_state | FUNC | hartree_fock_occupied_qubits | 75 |
| src/quantum/hartree_fock_reference_state.p | src.quantum.hartree_fock_reference_state | FUNC | bitstring_qn1_to_q0 | 103 |
| src/quantum/hartree_fock_reference_state.p | src.quantum.hartree_fock_reference_state | FUNC | hartree_fock_bitstring | 119 |
| src/quantum/hartree_fock_reference_state.p | src.quantum.hartree_fock_reference_state | FUNC | hartree_fock_statevector | 133 |
| src/quantum/hartree_fock_reference_state.p | src.quantum.hartree_fock_reference_state | FUNC | _half_filled_num_particles | 160 |
| src/quantum/hartree_fock_reference_state.p | src.quantum.hartree_fock_reference_state | FUNC | _phonon_vacuum_bitstring | 168 |
| src/quantum/hartree_fock_reference_state.p | src.quantum.hartree_fock_reference_state | FUNC | hubbard_holstein_reference_state | 190 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | CLASS | _FallbackLog | 44 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | CONST | SPIN_UP | 64 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | CONST | SPIN_DN | 65 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | CONST | LATEX_TERMS | 67 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | mode_index | 110 |

| path | module | kind | name | line |
|---|---|---|---|---|
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _prod | 141 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | n_sites_from_dims | 148 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | coord_to_site_index | 162 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | bravais_nearest_neighbor_edges | 178 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | jw_number_operator | 226 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _resolve_hubbard_nq | 253 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | boson_qubits_per_site | 268 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | phonon_qubit_indices_for_site | 288 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _normalize_pauli_symbol | 318 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _identity_pauli_polynomial | 331 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _pauli_monomial_global | 336 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _boson_local_matrices | 360 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _embed_to_qubit_space | 382 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _pauli_basis_mats | 399 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _matrix_to_pauli_coeffs | 423 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | boson_local_operator_pauli_decomp | 441 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | boson_operator | 457 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | boson_number_operator | 529 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | boson_displacement_operator | 550 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _unary_site_qubits_check | 593 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | boson_unary_number_operator | 602 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | boson_unary_displacement_operator | 640 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | boson_unary_bdag_operator | 674 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | boson_unary_b_operator | 716 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | build_hubbard_kinetic | 756 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | build_hubbard_onsite | 796 |

| path | module | kind | name | line |
|---|---|---|---|---|
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _parse_site_potential | 825 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | build_hubbard_potential | 846 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | build_hubbard_hamiltonian | 878 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | _eval_site_potential_maybe_time_dependent | 921 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | build_holstein_phonon_energy | 937 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | build_holstein_coupling | 983 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | build_hubbard_holstein_drive | 1038 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | build_hubbard_holstein_hamiltonian | 1068 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | show_latex_and_code | 1140 |
| src/quantum/hubbard_latex_python_pairs.py | src.quantum.hubbard_latex_python_pairs | FUNC | show_hubbard_latex_python_pairs | 1153 |
| src/quantum/pauli_polynomial_class.py | src.quantum.pauli_polynomial_class | CLASS | PauliPolynomial | 8 |
| src/quantum/pauli_polynomial_class.py | src.quantum.pauli_polynomial_class | CLASS | fermion_plus_operator | 156 |
| src/quantum/pauli_polynomial_class.py | src.quantum.pauli_polynomial_class | CLASS | fermion_minus_operator | 184 |
| src/quantum/qubitization_module.py | src.quantum.qubitization_module | CLASS | PauliTerm | 10 |
| src/quantum/time_propagation/cfqm_schemes | src.quantum.time_propagation.cfqm_scheme | CONST | _DEFAULT_TOL | 17 |
| src/quantum/time_propagation/cfqm_schemes | src.quantum.time_propagation.cfqm_scheme | CONST | _SCHEME_DATA | 20 |
| src/quantum/time_propagation/cfqm_schemes | src.quantum.time_propagation.cfqm_scheme | CONST | _SCHEME_ALIASES | 70 |
| src/quantum/time_propagation/cfqm_schemes | src.quantum.time_propagation.cfqm_scheme | FUNC | _norm_key | 80 |
| src/quantum/time_propagation/cfqm_schemes | src.quantum.time_propagation.cfqm_scheme | FUNC | _close | 84 |
| src/quantum/time_propagation/cfqm_schemes | src.quantum.time_propagation.cfqm_scheme | FUNC | _row_sums | 88 |
| src/quantum/time_propagation/cfqm_schemes | src.quantum.time_propagation.cfqm_scheme | FUNC | _col_sums | 92 |
| src/quantum/time_propagation/cfqm_schemes | src.quantum.time_propagation.cfqm_scheme | FUNC | validate_scheme | 103 |
| src/quantum/time_propagation/cfqm_schemes | src.quantum.time_propagation.cfqm_scheme | FUNC | get_cfqm_scheme | 170 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CLASS | _FallbackLog | 33 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CONST | LATEX_TERMS | 158 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | _normalize_pauli_string | 206 |

| path | module | kind | name | line |
|---|---|---|---|---|
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | basis_state | 221 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | apply_pauli_string | 237 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | expval_pauli_string | 274 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | expval_pauli_polynomial | 279 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | expval_pauli_polynomial_one_apply | 304 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | apply_pauli_rotation | 340 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | apply_exp_pauli_polynomial | 350 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | half_filled_num_particles | 388 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | jw_number_operator | 401 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | hubbard_hop_term | 422 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | hubbard_onsite_term | 437 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | hubbard_potential_term | 450 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | _parse_site_potential | 460 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | _single_term_polynomials_sorted | 492 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CLASS | AnsatzTerm | 523 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CLASS | HubbardTermwiseAnsatz | 530 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CLASS | HubbardLayerwiseAnsatz | 632 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CLASS | HardcodedUCCSDAnsatz | 717 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CLASS | HardcodedUCCSDLayerwiseAnsatz | 898 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CLASS | HubbardHolsteinTermwiseAnsatz | 1034 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CLASS | HubbardHolsteinPhysicalTermwiseAnsatz | 1214 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CLASS | HubbardHolsteinLayerwiseAnsatz | 1431 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CLASS | VQEResult | 1645 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | _try_import_scipy_minimize | 1655 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | vqe_minimize | 1663 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | CONST | _PAULI_MATS | 2011 |

| path | module | kind | name | line |
|---|---|---|---|---|
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | pauli_matrix | 2019 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | hamiltonian_matrix | 2027 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | _spin_orbital_index_sets | 2042 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | _sector_basis_indices | 2052 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | exact_ground_energy_sector | 2068 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | _sector_basis_indices_fermion_only | 2088 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | exact_ground_energy_sector_hh | 2109 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | show_latex_and_code | 2147 |
| src/quantum/vqe_latex_python_pairs.py | src.quantum.vqe_latex_python_pairs | FUNC | show_vqe_latex_python_pairs | 2159 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CONST | REPO_ROOT | 23 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestCompiledPauliCache | 65 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestAdaptCompiledStateBackendParity | 156 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestAdaptCLIParsing | 219 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestPoolBuilders | 231 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestPAOPPoolBuilder | 280 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestHHUCCSDPAOPCompositePoolBuilder | 450 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestSectorFilteringDispatch | 517 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestAdaptVQEHubbardUCCSD | 558 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestAdaptVQEHolsteinHVA | 668 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestAdaptVQEHolsteinPAOP | 749 |
| test/test_adapt_vqe_integration.py | test.test_adapt_vqe_integration | CLASS | TestAdaptEdgeCases | 878 |
| test/test_boson_unary.py | test.test_boson_unary | CONST | REPO_ROOT | 23 |
| test/test_boson_unary.py | test.test_boson_unary | FUNC | _term_strings | 52 |
| test/test_boson_unary.py | test.test_boson_unary | FUNC | _pauli_letters_in_poly | 57 |
| test/test_boson_unary.py | test.test_boson_unary | FUNC | _is_hermitian | 67 |
| test/test_boson_unary.py | test.test_boson_unary | FUNC | _one_hot_projector | 71 |

| path | module | kind | name | line |
|---|---|---|---|---|
| test/test_boson_unary.py | test.test_boson_unary | FUNC | _binary_physical_projector | 107 |
| test/test_boson_unary.py | test.test_boson_unary | CLASS | TestUnaryQPBAndMapping | 147 |
| test/test_boson_unary.py | test.test_boson_unary | CLASS | TestUnaryDisplacementXXYYOnly | 178 |
| test/test_boson_unary.py | test.test_boson_unary | CLASS | TestUnaryNumberIZOnly | 214 |
| test/test_boson_unary.py | test.test_boson_unary | CLASS | TestUnaryHermiticity | 252 |
| test/test_boson_unary.py | test.test_boson_unary | CLASS | TestUnaryFullHHBuilds | 301 |
| test/test_boson_unary.py | test.test_boson_unary | CLASS | TestBinaryVsUnarySpectrum | 331 |
| test/test_boson_unary.py | test.test_boson_unary | CLASS | TestUnaryReferenceStateVacuum | 390 |
| test/test_boson_unary.py | test.test_boson_unary | CLASS | TestUnaryFactoryDispatch | 460 |
| test/test_boson_unary.py | test.test_boson_unary | CLASS | TestUnaryBBdagAlgebra | 516 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | CONST | REPO_ROOT | 14 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | FUNC | _run_cfqm_total | 29 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | FUNC | _estimate_state_order_from_infidelity | 63 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | FUNC | test_cfqm4_static_regression_against_exact_ex | 80 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | FUNC | test_cfqm4_a0_safe_test_invariance_via_real_drive_ | 106 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | FUNC | test_manufactured_one_qubit_order_confirmat | 166 |
| test/test_cfqm_acceptance.py | test.test_cfqm_acceptance | FUNC | test_small_hh_cfqm4_converges_toward_fine_piecew | 231 |
| test/test_cfqm_propagator.py | test.test_cfqm_propagator | CONST | REPO_ROOT | 13 |
| test/test_cfqm_propagator.py | test.test_cfqm_propagator | CLASS | TestCFQMPropagator | 21 |
| test/test_cfqm_schemes.py | test.test_cfqm_schemes | CONST | REPO_ROOT | 6 |
| test/test_cfqm_schemes.py | test.test_cfqm_schemes | CLASS | TestCFQMSchemes | 13 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | CONST | REPO_ROOT | 23 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | FUNC | _sector_eigenvalues_pauli | 48 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | FUNC | _sector_eigenvalues_ed | 89 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | CONST | _ENCODINGS | 116 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | CONST | _PARAM_SETS | 119 |

| path | module | kind | name | line |
|---|---|---|---|---|
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | CLASS | TestEDvsPauliSpectrumMatch | 141 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | CLASS | TestEDHermiticity | 178 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | CLASS | TestEDBasisDimension | 206 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | CLASS | TestEDGroundStateEnergy | 229 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | CLASS | TestEncodingConsistency | 268 |
| test/test_ed_crosscheck.py | test.test_ed_crosscheck | CLASS | TestRandomizedCrosscheck | 304 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | CONST | REPO_ROOT | 26 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | FUNC | _build_L2_blocked_hamiltonian | 43 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | FUNC | _build_drive | 55 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | FUNC | _run_short_trajectory | 71 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | CLASS | TestDriveDisabledParity | 132 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | CLASS | TestDriveEnabledNonzeroAmplitude | 155 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | CLASS | TestDriveAmplitudeZeroSafeTest | 190 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | CLASS | TestSerializationKeys | 217 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | CLASS | TestBuildDriveMatrixAtTime | 240 |
| test/test_energy_total_observables.py | test.test_energy_total_observables | CLASS | TestTotalEnergyFormula | 294 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | CONST | REPO_ROOT | 38 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | FUNC | _import_evolve_piecewise_exact | 63 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | CLASS | TestReferenceMethodName | 73 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | FUNC | _exact_psi_sin_z | 124 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | FUNC | _run_piecewise_exact | 133 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | CLASS | TestPiecewiseExactConvergenceMidpoint | 160 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | CLASS | TestPiecewiseExactConvergenceLeft | 204 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | CLASS | TestJSONMetadataKeys | 240 |
| test/test_exact_steps_multiplier.py | test.test_exact_steps_multiplier | CLASS | TestCompareRunnerDefaults | 286 |
| test/test_hh_drive_qop_builder.py | test.test_hh_drive_qop_builder | FUNC | _coeff_map | 8 |

| path | module | kind | name | line |
|---|---|---|---|---|
| test/test_hh_drive_qop_builder.py | test.test_hh_drive_qop_builder | FUNC | test_drive_qop_builder_combines_static_and_drive_co | 15 |
| test/test_hh_drive_qop_builder.py | test.test_hh_drive_qop_builder | FUNC | test_drive_qop_builder_identity_fallback_when_all_terr | 28 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | FUNC | test_pauli_poly_to_sparse_pauli_op_preserves_exyz_to_ix | 22 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | FUNC | test_ansatz_to_circuit_matches_prepare_state_for_hh_termw | 37 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | FUNC | test_ideal_oracle_matches_statevector_expecta | 76 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | FUNC | test_shots_oracle_standard_error_improves_with_mor | 91 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | FUNC | test_aer_noise_mode_deterministic_with_fixed_seed_and_ | 127 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | FUNC | test_forced_aer_failure_triggers_sampler_fallba | 154 |
| test/test_hh_noise_oracle_runtime.py | test.test_hh_noise_oracle_runtime | FUNC | test_sampler_fallback_deterministic_with_fixed_s | 180 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | FUNC | test_parse_args_phase2_adapt_flags_presen | 16 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | FUNC | test_build_adapt_pool_hh_hva_nonempty | 43 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | FUNC | test_build_adapt_pool_hubbard_uccsd_nonemp | 51 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | FUNC | _legacy_ref_fixture | 70 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | FUNC | _new_traj_fixture | 81 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | FUNC | test_legacy_parity_exact_match_passes_strict_ | 100 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | FUNC | test_legacy_parity_tiny_mismatch_fails_strict_ | 113 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | FUNC | test_legacy_parity_time_grid_mismatch_fails_with_ | 126 |
| test/test_hh_noise_phase2_adapt.py | test.test_hh_noise_phase2_adapt | FUNC | test_legacy_comparison_plot_helper_smoke | 138 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | FUNC | test_hh_defaults_applied_from_minimum_table_l2 | 11 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | FUNC | test_hh_under_minimum_rejected_without_smoke | 21 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | FUNC | test_hh_under_minimum_allowed_with_smoke_ | 40 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | FUNC | test_hubbard_under_minimum_rejected_without_sm | 64 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | FUNC | test_cli_parses_fallback_flags | 87 |
| test/test_hh_noise_validation_cli.py | test.test_hh_noise_validation_cli | FUNC | test_cli_parses_legacy_parity_flags | 100 |
| test/test_hh_pool_b_union.py | test.test_hh_pool_b_union | FUNC | _single_term_op | 9 |
| test/test_hh_pool_b_union.py | test.test_hh_pool_b_union | FUNC | test_pool_b_strict_union_dedups_and_tracks_pres | 15 |

| path | module | kind | name | line |
|---|---|---|---|---|
| test/test_hh_seq_transition_logic.py | test.test_hh_seq_transition_logic | FUNC | test_windowed_slope_transition_triggers_after_pa | 11 |
| test/test_hh_seq_transition_logic.py | test.test_hh_seq_transition_logic | FUNC | test_windowed_slope_transition_does_not_trigger_when_st | 32 |
| test/test_hubbard_adapt_ref_source.py | test.test_hubbard_adapt_ref_source | CONST | REPO_ROOT | 13 |
| test/test_hubbard_adapt_ref_source.py | test.test_hubbard_adapt_ref_source | FUNC | _hh_state_dim | 22 |
| test/test_hubbard_adapt_ref_source.py | test.test_hubbard_adapt_ref_source | FUNC | _basis0 | 28 |
| test/test_hubbard_adapt_ref_source.py | test.test_hubbard_adapt_ref_source | CLASS | TestParseArgsAdaptRefSource | 34 |
| test/test_hubbard_adapt_ref_source.py | test.test_hubbard_adapt_ref_source | CLASS | TestAdaptRefSourceVQEPath | 56 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | FUNC | n_ph_max | 39 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | FUNC | L | 44 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | CLASS | TestHHReferenceStateShape | 52 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | CLASS | TestHHReferenceStateNorm | 77 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | CLASS | TestHHReferenceStateDeterminism | 93 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | CLASS | TestHHLayerwiseAnsatzParamCount | 106 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | CLASS | TestHHLayerwiseAnsatzNorm | 145 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | CLASS | TestHHLayerwiseAnsatzDeterminism | 172 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | CLASS | TestHubbardOnlyBackwardCompat | 191 |
| test/test_hubbard_holstein_ansatz.py | test.test_hubbard_holstein_ansatz | CLASS | TestHHReferencePhononVacuum | 225 |
| test/test_integration.py | test.test_integration | FUNC | _add_repo_paths | 13 |
| test/test_integration.py | test.test_integration | FUNC | _run_import_check | 22 |
| test/test_integration.py | test.test_integration | FUNC | _import_pauli_letters | 35 |
| test/test_integration.py | test.test_integration | FUNC | _import_pauli_term | 44 |
| test/test_integration.py | test.test_integration | FUNC | _import_pauli_polynomial | 50 |
| test/test_integration.py | test.test_integration | FUNC | _import_hubbard_helpers | 64 |
| test/test_integration.py | test.test_integration | FUNC | _import_hf_helpers | 78 |
| test/test_integration.py | test.test_integration | FUNC | main | 90 |
| test/test_pauli_polynomial_ops.py | test.test_pauli_polynomial_ops | CONST | REPO_ROOT | 10 |

| path | module | kind | name | line |
|---|---|---|---|---|
| test/test_pauli_polynomial_ops.py | test.test_pauli_polynomial_ops | CONST | REPO_ROOT_STR | 11 |
| test/test_pauli_polynomial_ops.py | test.test_pauli_polynomial_ops | FUNC | _coeff_map | 19 |
| test/test_pauli_polynomial_ops.py | test.test_pauli_polynomial_ops | CLASS | TestPauliPolynomialOps | 23 |
| test/test_time_potential_drive.py | test.test_time_potential_drive | CONST | REPO_ROOT | 22 |
| test/test_time_potential_drive.py | test.test_time_potential_drive | CLASS | TestZLabelPlacement | 35 |
| test/test_time_potential_drive.py | test.test_time_potential_drive | CLASS | TestDriveOnlyPhase | 73 |
| test/test_time_potential_drive.py | test.test_time_potential_drive | CLASS | TestQuadratureOrder | 157 |
| test/test_time_potential_drive.py | test.test_time_potential_drive | CLASS | TestPiecewiseExact | 205 |
| test/test_time_potential_drive.py | test.test_time_potential_drive | CLASS | TestWaveform | 246 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | REPO_ROOT | 28 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _L | 72 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _T | 73 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _U | 74 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _OMEGA0 | 75 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _G_EP | 76 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _N_PH_MAX | 77 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _BOSON_ENCODING | 78 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _BOUNDARY | 79 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _ORDERING | 80 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _NUM_PARTICLES | 81 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CONST | _ENCODINGS | 82 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | FUNC | _build_hh_poly_and_matrix | 85 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | FUNC | _build_hubbard_poly_and_matrix | 100 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CLASS | TestHHTrotterSmoke | 115 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CLASS | TestHHToHubbardReduction | 174 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CLASS | TestComparePipelineHHGuard | 215 |

| path | module | kind | name | line |
|---|---|---|---|---|
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CLASS | TestQiskitBaselineHHGuard | 247 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CLASS | TestPDFManifestFields | 275 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CLASS | TestSectorFilteringHH | 324 |
| test/test_trotter_hh_integration.py | test.test_trotter_hh_integration | CLASS | TestHHExactGroundEnergy | 379 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _L | 40 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _T | 41 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _U | 42 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _DV | 43 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _OMEGA0 | 44 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _G_EP | 45 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _N_PH_MAX | 46 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _BOSON_ENCODING | 47 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _BOUNDARY | 48 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _ORDERING | 49 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _HALF_FILL | 50 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CONST | _ENCODINGS | 51 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | FUNC | _build_hh_hamiltonian | 58 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | FUNC | _run_hh_vqe | 70 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | FUNC | _run_hubbard_vqe | 94 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CLASS | TestHHVQESmoke | 129 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CLASS | TestZeroCouplingLimit | 159 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CLASS | TestSeedReproducibility | 172 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CLASS | TestVQEProgressTelemetry | 195 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CLASS | TestBackwardCompat | 245 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CLASS | TestThetaEnergyConsistency | 279 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CLASS | TestVariationalBound | 308 |

| path | module | kind | name | line |
|---|---|---|---|---|
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | FUNC | _run_hh_vqe_termwise | 343 |
| test/test_vqe_hh_integration.py | test.test_vqe_hh_integration | CLASS | TestTermwiseAnsatz | 370 |

| path | flag | default | choices | line |
|---|---|---|---|---|
| elines/exact_bench/hh_noise_hardware_validat | --allow-aer-fallback | None | None | 1198 |
| elines/exact_bench/hh_noise_hardware_validat | --backend-name | None | None | 1195 |
| elines/exact_bench/hh_noise_hardware_validat | --compare-observables | energy_static_trotte | None | 1253 |
| elines/exact_bench/hh_noise_hardware_validat | --legacy-parity-tol | 1e-10 | None | 1251 |
| elines/exact_bench/hh_noise_hardware_validat | --legacy-reference-json | None | None | 1250 |
| elines/exact_bench/hh_noise_hardware_validat | --no-allow-aer-fallback | None | None | 1199 |
| elines/exact_bench/hh_noise_hardware_validat | --no-omp-shm-workaround | None | None | 1202 |
| elines/exact_bench/hh_noise_hardware_validat | --noise-mode | ideal | ['ideal', 'shots', 'aer_no | 1190 |
| elines/exact_bench/hh_noise_hardware_validat | --omp-shm-workaround | None | None | 1201 |
| elines/exact_bench/hh_noise_hardware_validat | --oracle-aggregate | mean | ['mean', 'median'] | 1194 |
| elines/exact_bench/hh_noise_hardware_validat | --oracle-repeats | 1 | None | 1193 |
| elines/exact_bench/hh_noise_hardware_validat | --output-compare-plot | None | None | 1252 |
| elines/exact_bench/hh_noise_hardware_validat | --shots | 2048 | None | 1191 |
| elines/exact_bench/hh_noise_hardware_validat | --use-fake-backend | None | None | 1196 |
| elines/exact_bench/hh_noise_hardware_validat | --vqe-method | None | ['SLSQP', 'COBYLA', 'L-BFG | 1217 |
| lines/exact_bench/hh_noise_robustness_seq_re | --allow-aer-fallback | None | None | 3515 |
| lines/exact_bench/hh_noise_robustness_seq_re | --backend-name | FakeManilaV2 | None | 3512 |
| lines/exact_bench/hh_noise_robustness_seq_re | --final-method | None | ['COBYLA', 'SLSQP', 'L-BFG | 3482 |
| lines/exact_bench/hh_noise_robustness_seq_re | --no-allow-aer-fallback | None | None | 3516 |
| lines/exact_bench/hh_noise_robustness_seq_re | --no-omp-shm-workaround | None | None | 3519 |
| lines/exact_bench/hh_noise_robustness_seq_re | --noise-modes | ideal,shots,aer_nois | None | 3505 |
| lines/exact_bench/hh_noise_robustness_seq_re | --noise-seed | 7 | None | 3511 |
| lines/exact_bench/hh_noise_robustness_seq_re | --noisy-methods | suzuki2,cfqm4 | None | 3506 |
| lines/exact_bench/hh_noise_robustness_seq_re | --noisy-mode-timeout-s | 1200 | None | 3520 |
| lines/exact_bench/hh_noise_robustness_seq_re | --omp-shm-workaround | None | None | 3518 |
| lines/exact_bench/hh_noise_robustness_seq_re | --oracle-aggregate | mean | ['mean', 'median'] | 3510 |

| path | flag | default | choices | line |
|---|---|---|---|---|
| lines/exact_bench/hh_noise_robustness_seq_re | --oracle-repeats | 4 | None | 3509 |
| lines/exact_bench/hh_noise_robustness_seq_re | --require-at-least-one-noisy | True | None | 3526 |
| lines/exact_bench/hh_noise_robustness_seq_re | --shots | 2048 | None | 3508 |
| lines/exact_bench/hh_noise_robustness_seq_re | --use-fake-backend | None | None | 3513 |
| lines/exact_bench/hh_noise_robustness_seq_re | --warm-method | None | ['COBYLA', 'SLSQP', 'L-BFG | 3457 |

SECTION: APPENDIX IMPORT EDGES | Scope-focused import edge appendix

| src | dst |
|---|---|
| pipelines.exact_bench.cfqm_vs_suzuki_qproc_proxy_benchmark | src.quantum.drives_time_potential |
| pipelines.exact_bench.cfqm_vs_suzuki_qproc_proxy_benchmark | src.quantum.time_propagation.cfqm_schemes |
| pipelines.exact_bench.cross_check_suite | src.quantum.hartree_fock_reference_state |
| pipelines.exact_bench.cross_check_suite | src.quantum.hubbard_latex_python_pairs |
| pipelines.exact_bench.cross_check_suite | src.quantum.pauli_polynomial_class |
| pipelines.exact_bench.cross_check_suite | src.quantum.vqe_latex_python_pairs |
| pipelines.exact_bench.hh_noise_hardware_validation | pipelines.exact_bench.noise_oracle_runtime |
| pipelines.exact_bench.hh_noise_hardware_validation | src.quantum.hartree_fock_reference_state |
| pipelines.exact_bench.hh_noise_hardware_validation | src.quantum.hubbard_latex_python_pairs |
| pipelines.exact_bench.hh_noise_hardware_validation | src.quantum.pauli_polynomial_class |
| pipelines.exact_bench.hh_noise_hardware_validation | src.quantum.qubitization_module |
| pipelines.exact_bench.hh_noise_hardware_validation | src.quantum.vqe_latex_python_pairs |
| pipelines.exact_bench.hh_noise_robustness_seq_report | pipelines.exact_bench.hh_seq_transition_utils |
| pipelines.exact_bench.hh_noise_robustness_seq_report | pipelines.exact_bench.noise_oracle_runtime |
| pipelines.exact_bench.hh_noise_robustness_seq_report | pipelines.hardcoded |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.drives_time_potential |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.hartree_fock_reference_state |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.hubbard_latex_python_pairs |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.operator_pools |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.pauli_polynomial_class |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.qubitization_module |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.time_propagation.cfqm_schemes |
| pipelines.exact_bench.hh_noise_robustness_seq_report | src.quantum.vqe_latex_python_pairs |
| pipelines.exact_bench.hh_seq_transition_utils | src.quantum.vqe_latex_python_pairs |
| pipelines.exact_bench.l3_hh_exp_fidelity_wrapper | src.quantum.hubbard_latex_python_pairs |
| pipelines.exact_bench.l3_hh_exp_fidelity_wrapper | src.quantum.operator_pools |

| src | dst |
|---|---|
| pipelines.exact_bench.l3_hh_exp_fidelity_wrapper | src.quantum.vqe_latex_python_pairs |
| pipelines.exact_bench.overnight_l3_hh_four_method_benchmark | src.quantum.hubbard_latex_python_pairs |
| pipelines.exact_bench.overnight_l3_hh_four_method_benchmark | src.quantum.operator_pools |
| pipelines.exact_bench.overnight_l3_hh_four_method_benchmark | src.quantum.vqe_latex_python_pairs |
| pipelines.hardcoded.adapt_pipeline | src.quantum.hartree_fock_reference_state |
| pipelines.hardcoded.adapt_pipeline | src.quantum.hubbard_latex_python_pairs |
| pipelines.hardcoded.adapt_pipeline | src.quantum.operator_pools |
| pipelines.hardcoded.adapt_pipeline | src.quantum.pauli_polynomial_class |
| pipelines.hardcoded.adapt_pipeline | src.quantum.vqe_latex_python_pairs |
| pipelines.hardcoded.bench_expm | src.quantum.drives_time_potential |
| pipelines.hardcoded.bench_expm | src.quantum.hubbard_latex_python_pairs |
| pipelines.hardcoded.hubbard_pipeline | pipelines.hardcoded |
| pipelines.hardcoded.hubbard_pipeline | src.quantum.drives_time_potential |
| pipelines.hardcoded.hubbard_pipeline | src.quantum.hartree_fock_reference_state |
| pipelines.hardcoded.hubbard_pipeline | src.quantum.hubbard_latex_python_pairs |
| pipelines.hardcoded.hubbard_pipeline | src.quantum.time_propagation.cfqm_schemes |
| pipelines.qiskit_archive.compare_hc_vs_qk | src.quantum.drives_time_potential |
| pipelines.qiskit_archive.qiskit_baseline | src.quantum.drives_time_potential |
| pipelines.qiskit_archive.qiskit_baseline | src.quantum.hartree_fock_reference_state |
| src.quantum.drives_time_potential | src.quantum.hubbard_latex_python_pairs |
| src.quantum.ed_hubbard_holstein | src.quantum.hubbard_latex_python_pairs |
| src.quantum.hartree_fock_reference_state | src.quantum.hubbard_latex_python_pairs |
| src.quantum.hubbard_latex_python_pairs | src.quantum.pauli_polynomial_class |
| src.quantum.hubbard_latex_python_pairs | src.quantum.qubitization_module |
| src.quantum.operator_pools.polaron_paop | src.quantum.hubbard_latex_python_pairs |
| src.quantum.operator_pools.polaron_paop | src.quantum.pauli_polynomial_class |

| src | dst |
|---|---|
| src.quantum.operator_pools.polaron_paop | src.quantum.qubitization_module |
| src.quantum.pauli_polynomial_class | src.quantum.qubitization_module |
| src.quantum.pauli_words | src.quantum.qubitization_module |
| src.quantum.qubitization_module | src.quantum.pauli_letters_module |
| src.quantum.vqe_latex_python_pairs | hartree_fock_reference_state |
| src.quantum.vqe_latex_python_pairs | src.quantum.compiled_polynomial |
| src.quantum.vqe_latex_python_pairs | src.quantum.hartree_fock_reference_state |
| src.quantum.vqe_latex_python_pairs | src.quantum.hubbard_latex_python_pairs |
| src.quantum.vqe_latex_python_pairs | src.quantum.pauli_polynomial_class |
| src.quantum.vqe_latex_python_pairs | src.quantum.pauli_words |
| test.test_adapt_vqe_integration | src.quantum.hubbard_latex_python_pairs |
| test.test_adapt_vqe_integration | src.quantum.operator_pools |
| test.test_adapt_vqe_integration | src.quantum.vqe_latex_python_pairs |
| test.test_boson_unary | src.quantum.hartree_fock_reference_state |
| test.test_boson_unary | src.quantum.hubbard_latex_python_pairs |
| test.test_boson_unary | src.quantum.vqe_latex_python_pairs |
| test.test_cfqm_acceptance | pipelines.hardcoded.hubbard_pipeline |
| test.test_cfqm_acceptance | src.quantum.drives_time_potential |
| test.test_cfqm_acceptance | src.quantum.hubbard_latex_python_pairs |
| test.test_cfqm_acceptance | src.quantum.time_propagation.cfqm_propagator |
| test.test_cfqm_acceptance | src.quantum.time_propagation.cfqm_schemes |
| test.test_cfqm_propagator | src.quantum.time_propagation.cfqm_propagator |
| test.test_cfqm_propagator | src.quantum.time_propagation.cfqm_schemes |
| test.test_cfqm_schemes | src.quantum.time_propagation.cfqm_schemes |
| test.test_ed_crosscheck | src.quantum.ed_hubbard_holstein |
| test.test_ed_crosscheck | src.quantum.hubbard_latex_python_pairs |

| src | dst |
|---|---|
| test.test_ed_crosscheck | src.quantum.vqe_latex_python_pairs |
| test.test_energy_total_observables | pipelines.hardcoded.hubbard_pipeline |
| test.test_energy_total_observables | src.quantum.drives_time_potential |
| test.test_energy_total_observables | src.quantum.hartree_fock_reference_state |
| test.test_energy_total_observables | src.quantum.hubbard_latex_python_pairs |
| test.test_exact_steps_multiplier | pipelines.hardcoded.hubbard_pipeline |
| test.test_exact_steps_multiplier | pipelines.qiskit_archive.compare_hc_vs_qk |
| test.test_exact_steps_multiplier | src.quantum.drives_time_potential |
| test.test_hh_drive_qop_builder | pipelines.exact_bench.hh_seq_transition_utils |
| test.test_hh_noise_oracle_runtime | pipelines.exact_bench.noise_oracle_runtime |
| test.test_hh_noise_oracle_runtime | src.quantum.hartree_fock_reference_state |
| test.test_hh_noise_oracle_runtime | src.quantum.pauli_polynomial_class |
| test.test_hh_noise_oracle_runtime | src.quantum.qubitization_module |
| test.test_hh_noise_oracle_runtime | src.quantum.vqe_latex_python_pairs |
| test.test_hh_noise_phase2_adapt | pipelines.exact_bench.hh_noise_hardware_validation |
| test.test_hh_noise_validation_cli | pipelines.exact_bench.hh_noise_hardware_validation |
| test.test_hh_pool_b_union | pipelines.exact_bench.hh_seq_transition_utils |
| test.test_hh_pool_b_union | src.quantum.pauli_polynomial_class |
| test.test_hh_pool_b_union | src.quantum.qubitization_module |
| test.test_hh_pool_b_union | src.quantum.vqe_latex_python_pairs |
| test.test_hh_seq_transition_logic | pipelines.exact_bench.hh_seq_transition_utils |
| test.test_hubbard_adapt_ref_source | pipelines.hardcoded.hubbard_pipeline |
| test.test_hubbard_adapt_ref_source | src.quantum.hubbard_latex_python_pairs |
| test.test_hubbard_holstein_ansatz | src.quantum.hartree_fock_reference_state |
| test.test_hubbard_holstein_ansatz | src.quantum.hubbard_latex_python_pairs |
| test.test_hubbard_holstein_ansatz | src.quantum.vqe_latex_python_pairs |

| src | dst |
|---|---|
| test.test_integration | src.quantum.hartree_fock_reference_state |
| test.test_integration | src.quantum.hubbard_latex_python_pairs |
| test.test_integration | src.quantum.pauli_letters_module |
| test.test_integration | src.quantum.pauli_polynomial_class |
| test.test_integration | src.quantum.pauli_words |
| test.test_pauli_polynomial_ops | src.quantum.pauli_polynomial_class |
| test.test_pauli_polynomial_ops | src.quantum.qubitization_module |
| test.test_time_potential_drive | pipelines.hardcoded.hubbard_pipeline |
| test.test_time_potential_drive | src.quantum.drives_time_potential |
| test.test_time_potential_drive | src.quantum.hubbard_latex_python_pairs |
| test.test_trotter_hh_integration | pipelines.hardcoded.hubbard_pipeline |
| test.test_trotter_hh_integration | pipelines.qiskit_archive.compare_hc_vs_qk |
| test.test_trotter_hh_integration | pipelines.qiskit_archive.qiskit_baseline |
| test.test_trotter_hh_integration | src.quantum.hartree_fock_reference_state |
| test.test_trotter_hh_integration | src.quantum.hubbard_latex_python_pairs |
| test.test_trotter_hh_integration | src.quantum.vqe_latex_python_pairs |
| test.test_vqe_hh_integration | src.quantum.hartree_fock_reference_state |
| test.test_vqe_hh_integration | src.quantum.hubbard_latex_python_pairs |
| test.test_vqe_hh_integration | src.quantum.vqe_latex_python_pairs |

```
SECTION: COMMAND + PROVENANCE

generated_at_utc: 2026-03-04T18:22:39.786027+00:00
run_command: /opt/anaconda3/bin/python /Users/jakestrobel/Documents/Holstein_implementation/Holstein_test/pipelines
    /exact_bench/hh_noise_model_repo_guide.py --output-json /Users/jakestrobel/Documents/Holstein_implementation/Ho
    lstein_test/artifacts/json/hh_noise_model_repo_guide_index.json --output-pdf
    /Users/jakestrobel/Documents/Holstein_implementation/Holstein_test/docs/HH_noise_model_repo_guide.pdf
summary_json:
    /Users/jakestrobel/Documents/Holstein_implementation/Holstein_test/docs/repo_guide_assets/repo_guide_summary.js
    on
summary_found: True
scope_module_count: 34
symbol_count: 418
cli_flag_count: 31
edge_count: 123
doc_ref_count: 338
test_anchor_count: 67

Required section markers included:
  - SECTION: SCOPE + SOURCES
  - SECTION: ARCHITECTURE MAP
  - SECTION: ORACLE CONTRACT
  - SECTION: VALIDATION PIPELINE CONTRACT
  - SECTION: ROBUSTNESS REPORT CONTRACT
  - SECTION: TEST + DOC COVERAGE
  - SECTION: REFACTOR PLAYBOOK
  - SECTION: APPENDIX SYMBOL INDEX
  - SECTION: APPENDIX CLI FLAG INDEX
  - SECTION: APPENDIX IMPORT EDGES
  - SECTION: COMMAND + PROVENANCE
```